Multiple Linear Regression - I

Hello, and welcome to another session on Business Statistics for Entrepreneurs. In the last module, we discussed simple linear regression, where a dependent variable y was a function of only a single independent variable x. In this module, we are going to continue our journey on statistical inference and look at multiple linear regression,

where this dependent variable y will be permitted to be a function of multiple independent variables x. Joining me in this journey are our bright students, Tejas and Ashwini. Hi. Along with them, I will cover some important concepts and relevant applications. As always, they are going to keep me honest and on track.

Welcome back, Tejas and Ashwini. Are you ready for multiple linear regression? Yes, absolutely. I cannot wait to go back to the examples we had in the last module and see what multiple linear regression will tell us. Yes, the same sample data we had in module 1 to understand descriptive statistics are now helping us form inferences about the population.

Those numbers have a lot of stories to tell. Yes, and like in the previous module with multiple linear regression, you will see a generalization of the simple linear regression concepts and also the applications of these concepts like confidence intervals and hypothesis test. So, what is our plan? Let me start with an overview of what we are going to be covering in this module.

We will start with the basic idea behind multiple linear regression, where we try to establish a linear relationship between a dependent variable y and several independent variables x. Like we did with simple linear regression, we will see the basic multiple linear regression model and the associated equation.

We will also see the estimated multiple regression equation. Then, we will walk through the steps that are typically used to build and validate and implement a multiple regression model. Then we will see an extension of the least square method to determine the coefficients for a multiple regression equation.

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We will also see a generalized version of the coefficient of determination known as the multiple coefficient of determination. We will contrast the assumptions that we made in simple linear regression with those that we will need to make in validating a multiple linear regression model. We will look at test for significance.

We will see an F test and we will see a T test. But we will also see a concept known as multicollinearity that comes when we have multiple independent variables. Then we will see how the estimated regression equation can be used for estimation and prediction. We will also see how to analyze the residuals to detect outliers and influential observations, just as we did with simple linear regression.

We will also see how this analysis can be used in the context of several applications. Let's get started. In the previous module, we learned about simple linear regression model. We saw how the model was used to develop an estimated regression equation that describes the relationship between two variables x and y.

If you remember, the variable being predicted or explained was known as the dependent variable. We called it y. The variable being used to predict or explain the dependent variable y was known as the independent variable. We denoted it by x. We had one dependent variable y and one independent variable x in that case.

In this module, we will continue our study of regression for situations that involve predicting a single dependent variable y, but using two or more independent variables. This is the essential difference between multiple linear regression and the simple linear regression that we saw. By this extension, we are now able to consider more factors that impact the dependent variable and this allows us to obtain better predictions than what would have been possible with just simple linear regression.

If you remember in some of the examples we did in the last module we could have made far better predictions it appeared using multiple linear regression than we were able to do with simple linear regression. As I just said, in multiple linear regression or MLR, we will study how a dependent variable y is related to two or more independent variables.

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From a notation perspective, we will use the letter P to denote the number of independent variables. This means that the equation that describes how the dependent variable y is related to the independent variables x1, x2 and so on up to xp and an error term will be called as our multiple regression model.

Like we did with simple linear regression, we will assume that this relationship between y and the dependent variables takes the following forms, that y is equal to beta 0 plus beta 1 times x 1 plus beta 2 times x 2 and so on until beta p times x p plus epsilon. In this model, beta 0, beta 1, beta 2 and so on until beta p are the parameters.

And the error term epsilon is a random variable. We can see that in this model too, y is a linear function, but now a function of x 1, x 2 and so on up to x p plus an error term epsilon. Now, this error term epsilon accounts for the variability in y that cannot be explained by the linear combination of x1, x2 all the way up to xp.

Like we did with simple linear regression, we will assume that the mean or the expected value of the random variable of epsilon is equal to 0. As a result of this assumption, the mean or the expected value of y denoted by E of y will be equal to beta 0, plus beta 1 times x 1 plus beta 2 times x 2 and so on up to beta p times x p.

This equation that describes how the mean value of y is related to the independent variables x 1, x 2 and so on up to x p is called as the multiple linear regression equation. So, formally we will say that the expectation of y, E of y, is equal to beta 0 plus beta 1 times x 1 plus beta 2 times x 2 and so on up to beta p times x p.

Of course, if the values of beta 0, beta 1, beta 2 and so on up to beta p were known, then we could use the multiple linear regression equation that we just saw to compute the mean value of y, E of y for particular values of x1, x2 and so on up to xp. But unfortunately these parameter values are usually not known and they must be determined using a sample data or training data.

Usually when we collect a simple random sample of data and use that as a training set, what we get are sample statistics b0, b1, b2 and so on up to bp, that are then used as point estimators of the unknown parameters beta 0, beta 1, beta 2 and so on up to beta p. These sample statistics

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then provide the estimated multiple linear regression equation, which is given by y hat is equal to b 0 plus b 1 times x 1 plus b 2 times x 2 and so on up to b p times x p, where b 0, b 1, b 2, b 3 and so on up to b p are the estimates of beta 0, beta 1, beta 2 and so on up to beta p. y hat is then the predicted value of the dependent variable. This overall estimation process is described in this figure.

Let us now look at the steps involved in developing a multiple linear regression model. These steps are similar to what we used to develop a simple linear regression model. However, there are some additional features such as multicollinearity, overfitting, etc. that have to be taken into account while performing multiple linear regression.

Let us look at some of the steps that are involved in the development of a multiple linear regression model. We should realize though that all model development is an iterative process and many of the steps that I am going to walk through in a few minutes might have to be repeated before we find the best multiple linear regression model required to answer questions to a problem.

Let's take a look at each of these steps. The first step in building a regression model is to collect and extract the relevant data for the problem identified. In the case of multiple linear regression models, we will have several independent variables and the required data could be collected from secondary sources such as an ERP system or a company database or government census data, as the case may be, or through primary sources such as surveys and interviews.

The second step is to pre process this data. In pre processing, we try to gauge the quality of the data in terms of its completeness and correctness. If the data that we are planning to use has a significant number of missing variables or observations, it might be hard to build credible models. We may also have to deal with qualitative variables

and categorical variables. What do we do with qualitative variables? With qualitative variables, we may need to do some variable transformation. We will see that later in this module. The third step is to perform descriptive analytics. As we have seen, there is always a good practice before we start to build models.

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By computing basic statistics and using data visualization tools like scatter plots and box plots. We can check the correlation between different variables. We can also provide some warnings about multicollinearity and overfitting issues that we might face. Descriptive statistics can also allow us to identify proxy variables in case the original variable cannot be used in the model for either ethical or regulatory reasons.

The fourth step is to decide the modeling strategy. Now, what we mean here is to decide what are the right set of independent variables that need to be used? The selected variables must exhibit certain desirable properties like they must be independent variables and should exhibit consistent model performance across different training and test data sets.

Also, the chosen independent variables must be controllable by the decision maker. What do you mean by controllable variable? Can you give an example? Yes, sure. So let's say as an example, sales in a store might be influenced by price and weather patterns. Price is a controllable variable, but weather is not, and a model, having weather as a variable might not be very useful.

Okay. The fifth step is now to divide the data into training and validation data. We would want to build multiple subsets of the data to validate the model multiple times. The sixth step is to define the functional form of the multiple linear regression equation. Usually, we start with a simple linear relationship between the dependent and the independent variables.

But sometimes we might need to change this functional form to better fit the data. We saw some examples of changed functional forms in simple linear regression as well. Do you remember? Yes, we saw that replacing x by x square does not make the regression non linear. Yes, exactly. Now, the seventh step is to estimate the regression parameters once this functional form has been specified.

Here, we can use techniques such as ordinary least squares to determine the coefficients such that the sum of square distance between the actual observations in the sample and the regression equation is minimized. The ordinary least square method actually provides the best linear unbiased estimator of the regression coefficients.

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The eighth step is to perform the regression model diagnostics. Here we first check for the statistical significance of the predictor variables. In addition, validating multiple linear regression model also involves for checking for issues such as the normality of residuals, multicollinearity, and heteroscedasticity.

All of these must be done in addition to checking for the normality of the residuals as we did with simple linear regression. Are we going to do similar hypothesis test as we did in SLR? Yes, along similar lines, we will use an F test to check for the overall significance of the model. And we will use a t test to test for the significance of the individual variables that are included in the model. Remedial measures must be taken if some of the model assumptions are violated. The ninth step includes validating the model using the validation data.

The final model selection would depend on the performance of the model on the validation data. Do we use coefficient of determination again? Yes, the common measures used to validate the model are R square and the adjusted R square, the mean absolute percentage error and the root mean square error measures.

Models that give consistent performance and by that we mean low bias and low variance in both the training and in the validation data set are usually chosen for deployment. The tenth and the final step is to deploy the model so that we can generate actionable items and an implementation plan. This overall process of developing a multiple linear regression model can be an iterative process and we may have to go through some of these steps multiple times to converge on the final model.

We should also monitor the performance of the model as it might need to be changed during the course of implementation. Now that we have seen the steps in model development, let us get into the details and let us start by looking at the least squares method that we use.

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MLR – OLS

When we discuss simple linear regression, we use the least squares method to develop an equation that best approximates the straight line relationship between the dependent and the independent variables. The same approach is used to develop the estimated multiple linear regression equation. The least squares criteria that we use is as follows. We want to minimize the sum of y i minus y i hat squared, where the summation is over all the observations.

Here, y i is the observed value of the dependent variable for the ith observation and y i hat is the predicted value of the dependent variable for the ith observation. The predicted values of the dependent variable are then computed using the estimated multiple regression equation called y hat is equal to b 0 plus b 1 times x 1 plus b 2 times x 2 and so on until b p times x p.

As you can see, the least square method uses the sample data to provide values of b0, b1, b2 and so on up to bp such that the sum of the squared residuals, the deviations between the observed values of the dependent variable yi and the predicted value of the dependent variable yi hat are minimized. In the case of simple linear regression, we were able to compute closed form expressions for b0 and b1 respectively.

However, in the case of multiple linear regression, the formulas for these regression coefficients b0, b1 and so on up to bp involve matrix algebra and we will not get into those details here. However, we will use Excel to compute these coefficients b0, b1 and so on and once we compute these coefficients, our emphasis will be on how to interpret the output rather than how to make these computations. To see how the coefficients of multiple linear regression are computed, let us return back to Hanumantha's example where he had data of his customers from Canara Bank. How many observations did he have, Tejas and Ashwini? In his data set, he had 55 observations.

Yes, and he had collected data about the monthly spend on credit cards, the annual income of the household, and the size of the household. In the previous module, we developed simple linear regression models to see the relationship between the monthly spend on the credit card and the annual income. We also developed a simple linear regression to see the relationship between monthly spend and the size of the household.

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Ashwini and Tejas, what do you remember about those simple linear regression models? The simple linear regression model between monthly spend and household income was much stronger than the model we got for simple linear regression between monthly spend and household size. Yes, that is correct. However, our analysis has shown that both linear relationships are significant.

Yes, Ashwini, that was also true. So let us remind ourselves about the details of these models. For the linear regression model between monthly spend y versus the annual income x, we had a hypothesis that the simple linear relationship will be of the form y is equal to beta 0 plus beta 1 times x 1 plus epsilon.

And we said that this could be used to describe the relationship between the monthly spend y and the independent variable annual income x 1. To estimate the parameters of beta 0 and beta 1, we use the least squares method. And to develop the estimated regression equation, y hat is equal to b0 plus b1 times x1.

Using Excel, we found out the estimated regression equation. Do you remember what it was? It was y hat is equal to 1047. 37 plus 85. 27 times x1. We also computed the f value for our f test in the ANOVA table. And what did we get our test statistic to be? It was equal to 98.24. Yes, and the corresponding p value was equal to zero.

This indicated that the relationship is significant. This finding is fairly good, but Hanumantha can perhaps do better by considering a second independent variable to explain some of the remaining variability in the dependent variable. Yes, because monthly spend and the size of the household also indicated a significant relationship.

Yes, and what did we find? We started with a model where monthly spend was y and the household size was x1. To estimate the parameters of beta 0 and beta 1, we use the least squares method again. And we use that to develop the estimated regression equation y hat is equal to b 0 plus b 1 times x 1. Using Excel, we found out the Estimated Regression Equation.

What did we get? We got y hat is equal to 2083. 67 plus 520. 13 times x1. Yes, and we also did the F test in the ANOVA table. And what did we get the value of the test statistic to be? It

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was equal to 34. 76. Yes, and the corresponding p value was fairly close to 0. So this indicated that the relationship is significant.

That is, we can reject the null hypothesis that beta 1 is equal to 0, because the p value was less than alpha of 0. 05. The same conclusion was obtained from the t test. And what was the test statistic there? It was a T random variable with a value of 5. 89. Yes, and that gave us a p value that was fairly close to zero.

Thus, we could conclude that the relationship between the total monthly spent and the annual income was indeed significant. And we also found that the coefficient of determination, R squared, as a percentage was 39%. Now, let us try and improve our regression models. Let us try a multiple linear regression model where we have two independent variables.

The first corresponding to annual income denoted by x1 and the second corresponding to the size of the household denoted by x2. Then the estimated regression equation would be y hat is equal to b0 plus b1 times x1 plus b2 times x2. Let us now use Excel's analysis tool pack to compute these coefficients.

Here I have a spreadsheet with the monthly spent in column D, the annual income in column B, and the household size in column C. We open the regression option in the analysis tool pack. And in the analysis tool pack, we select the Y variable to be the data in column D. And the X variables are going to be both the annual income and the household size.

There is the data in both columns B and C. In addition to obtaining the numerical output, let us also compute the residuals, the standard residuals and the plot of all the residuals. Did you get all of that? Give me a minute. Okay, got it. Okay, now let's look at them closely. The Analysis Toolpack output shows us that the estimated regression equation is given by y hat is equal to 355. 68 plus 71. 16 times the annual income x1 plus 339. 9 times the household size x2. That is, the value of b0 is equal to 335. 68, b1 is equal to 71. 16, and b2 is equal to 339. 9. The R square value is 0. 8 or 80 percent, which is far better than the R square that we got using only a single variable.

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Let us now discuss the interpretation of the coefficients in the estimated regression equation with only annual income as an independent variable versus the interpretation of those same coefficients in an estimated regression equation that includes both the annual income and the household size as the two independent variables.

We see that the value of b1 is not the same in both cases. Yes, in the simple linear regression with only annual income as the independent variable, the value of b1 was 85. 27. Exactly, whereas in the multiple linear regression equation with annual income and household size as the independent variables, the coefficient corresponding to annual income b1 is now 71.16. In simple linear regression, we interpreted b1 as the estimate of change in y for one unit change in the independent variable. That is, we said that for a one unit or a one lakh change in the annual income, b1 is equal to 85. 27 reflects the amount of change in the monthly spent. This interpretation needs to be modified a little bit in the context of multiple linear regression.

In multiple linear regression, we interpret the regression coefficient b i for the ith independent variable as the following. We say that b i represents an estimated change in the y corresponding to one unit change in the x i when all other independent variables are held constant. Therefore, in Hanumantha's case, we would say when we have two independent variables x1 and x2, that the interpretation of b1 equal to 71. 16 corresponding to the annual income would be as follows. We would say that every increase in annual income by 1 lakh is associated with an increase in monthly spend by 71. 16. Assuming that all other independent variables such as the household size in this case are held constant.

Similarly, the annual income remaining constant, the regression coefficient of B2 equal to 339.9 indicates that the increase in monthly spend for one unit increase in the household size is equal to 339. 9 when the annual income in lakhs is held to be constant.

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MLR - Coefficient of Determination

Next, let us look at the multiple coefficient of determination. This coefficient measures how good a fit is provided by the estimated regression equation y hat is equal to b 0 plus b 1 times x 1 plus b 2 times x 2 all the way up to b p times x p. In the case of simple linear regression, we saw that the total sum of squares SST can be partitioned into two components.

The sum of squares due to regression SSR and the sum of squares due to error SSE. The same procedure applies in the case of multiple linear regression. Let us now see the relationship between SST, SSR and SSE. Even in the case of multiple linear regression, SST is equal to SSR plus SSE. Here, SST is defined as the sum or the total sum of squares and it is computed as the sum of y i minus y bar square over all values of y, where y bar is the average of the y values y i corresponds to the ith observation.

SSR is the sum of squares due to regression and it is computed as the sum of y i hat minus y bar square over all the observations. Here y i hat is the estimate of the ith observation obtained from the regression and y bar is the average of all the y values. SSE is the sum of squares due to error and that is given by the sum of y i minus y i hat square over all the observations, where y i is the actual observation and y i hat is the estimate of the ith observation obtained from the multiple linear regression.

Because it is a mess to write the closed form expressions for the three sum of squares, we will not look at the detailed equations and the manual computation of these values. Instead, we will just interpret the output that we have obtained from the analysis tool pack. We see that the values in the table corresponding to the analysis of variance or ANOVA are part of the output.

We see that the sum of squares total or SST is equal to 1 1 4 1 9 7 9 2 3, and the sum of squares error SSE is equal to 2 2 7 2 9 0 7 0, and the sum of squares regression SSR is equal to 9 1 4 6 8 8 5 2. If we compare these values with the values that we obtained in simple linear regression with only annual income as the independent variable, we find that SST is equal to 114197923 and the sum of squares error SSE is equal to 40017408 and the sum of squares regression Is given by SSR equal to 7 4 1 8 0 5 1 6. What do you notice? It looks like the value of SST did

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not change. Yes, because the value of SST is equal to 1 1 4 1 9 7 9 2 3 in both cases, and it is the sum of the overall variation in the Y data.

SST then doesn't depend on y hat, the estimated value of y. Now, what about SSR? SSR increases when we go from one independent variable to two independent variables. Yes, it increases from 74180516 to 91468852. And what about SSE? SSE decreases when the second independent variable corresponding to household size is added.

Yes, it goes down from 4 0 0 1 7 4 0 8 to 2 2 7 2 9 0 7 0. The reason is that by adding independent variables, the estimated multiple linear regression provides a better fit for the observed data. The two variables are collectively able to explain a lot more of the variability observed in the Y values in the data.

Now, in the case of simple linear regression, we use the coefficient of determination. Do you remember what that was? Yes, R square to be equal to SSR divided by SST. Yes, and we used R square to measure the goodness of fit for the estimated regression equation. The same concept will apply in multiple linear regression as well.

In multiple linear regression, we will use the term multiple coefficient of determination to measure the goodness of fit of the estimated multiple linear regression equation. Do we denote it by a different notation? No Tejas, the term also is denoted by R square, but the multiple linear coefficient of determination R square is also defined as SSR over SST.

And it can be interpreted as a proportion of the variability in the dependent variables that can be explained by the estimated multiple regression equation. And if you multiply the r square by 100, it can be interpreted as a percentage of variability in the y values that is explained by the estimated regression equation.

So in Hanumanatha's example, we have SSR for the case with two independent variables to be equal to 9 1 4 6 8 8 5 2. And we have SST to be equal to 1 1 4 1 9 7 9 2 3, which gives us an R square of 9 1 4 6 8 8 5 2 divided by 1 1 4 1 9 7 9 2 3. That comes out to be around 0. 80. Does that mean that adding independent variable causes the prediction errors to become smaller and thus reducing the sum of squares error?

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Yes Tejas, that's right. And because SSR is equal to SST minus SSE, when SSE becomes smaller, SSR becomes larger causing the R square that is equal to SSR divided by SST to go up. Now, if we convert this R square into a percentage, we can see that 80 percent of the variability in monthly spend is explained by the estimated multiple linear regression equation that has both the annual income and the household size as independent variables.

Notice that this R square increased from the value of 0. 65 that we had when we had only one independent variable corresponding to monthly income. Thus, the percentage of variability in annual spend that is explained by this estimated regression equation increases from 65 percent to 80 percent when the household size is added as a second independent variable.

Will R square always increase as you add more independent variables to the model? Yes, Ashwini, it usually does. In the output, we also see another value called adjusted R square. What is that value? That value has a connection to the previous question that you just asked. Many people prefer to adjust the R square value that we just computed as a ratio of SSR over SST to account for the number of independent variables.

So, this helps to avoid overestimating the impact of adding independent variables on the amount of variability that is explained by the estimated regression equation. How is adjusted R square different from R square? If n is the number of observations that we have and p is the number of independent variables then the adjusted multiple coefficient of determination and R square are related as follows.

The adjusted R square is given by 1 minus the product of 1 minus R square times n minus 1 divided by n minus p minus 1. So in Hanumantha's example, we have n equal to 55. In the case of the simple linear regression, the number of independent variables p is equal to 1. So, the adjusted R square would be 1 minus 1 minus R square times 55 minus 1 divided by 55 minus 2.

In the simple linear regression case, we had found the R square to be equal to 0. 65. So, the adjusted R square in that case would be 1 minus 0. 35 times 54 divided by 53 and that comes out to 0. 64. In the case of multiple linear regression, we had found the R square to be equal to

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0. 8. We have n equal to 55 again, but now the number of independent variables p is equal to 2.

So the adjusted R square will be equal to 1 minus 1 minus 0. 8 times 54 divided by 52 and that comes out to be equal to 0. 793. So, therefore, after adjusting for the two independent variables, we see that we have an adjusted multiple coefficient of determination, which is 0. 79 or 79 percent. The adjusted R square did not change much.

Is that usually the case? Yes, that is usually the case. But if we have a large value of p, then n minus 1 divided by n minus p minus 1 could be negative, at least technically, giving negative R square. Yes, technically you are right. And for that to happen, p must be more than n, lots of independent variables with very few observations.

And if that happens, and when that happens, we just set the adjusted R square value to be equal to 0. We can still use R square. Yes, we can still use R square in that case.

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MLR - Test of Significance

Now, let us recap what we have done so far. For the case of multiple linear regression, the assumed regression equation is y is equal to beta 0 plus beta 1 times x 1 plus beta 2 times x 2 and so on till beta p times x p plus an epsilon. Then, the least square method was used to develop or determine b0, b1, b2, and so on up to b.

These were the estimates of the model parameters beta 0, beta 1, and so on up to beta p. Using those, we developed an estimated regression equation to be y hat is equal to b0 plus b1 times x1 plus b2 times x2 and so on up to bp times xp. Then, using the value of the coefficient of determination, R square, we measured the goodness of fit of the estimated regression equation.

We applied these concepts to Hanumantha's data set using two independent variables that improved our R square from 0. 65 to 0. 80. However, even if the value of R square is high, this estimated regression equation should not be used for decision making until the appropriateness of the assumed model has been evaluated.

This involves testing the significance of linear relationships that we are assuming in the model. When we test the significance of the linear relationship, we make a few assumptions about the error term epsilon. Let us see those assumptions and they are similar to what we had in simple linear regression.

The first assumption is that the error term epsilon is a random variable with a mean or expected value of 0. The expected value of y, E of y, will be equal to beta 0 plus beta 1 times x 1 plus beta 2 times x 2 and so on up to beta p times x p. E of y is the average of all possible values of y that might occur for the given values of x1, x2 and so on up to xp.

The second assumption is that the variance of epsilon, the error term denoted by sigma square, is the same for all values of x1, x2 and so on up to xp. This means that the variance of y about the regression line equals sigma square and is the same for all values of x1, x2 and so on up to xp. The third assumption is that the values of the error term epsilon are all independent.

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Now since the value of epsilon the error term for a particular set of values of x1, x2 and so on up to xp are not related to the values of epsilon for another set of values x1, x2 and so on up to xp, then the value of y for a particular set of x1, x2 and so on up to xp is also not related to the value of y for another set of values of x1, x2 and so on up to xp.

The fourth assumption is that the error term epsilon is normally distributed for all values of x1, x2 and so on up to xp. This assumption along with our earlier assumptions imply that we are assuming that the error term epsilon is normally distributed with mean 0 and variance equal to sigma square.

Now, because y is a linear function of epsilon, y is also normally distributed random variables for all values of x. To obtain more insights about the form of the relationship given by the multiple linear regression equation, let us consider the following case of two independent variables, where the expectation of y is given by beta 0 plus beta 1 times x 1 plus beta 2 times x 2.

The graph of this equation would be a plane in the three dimensional space. You can see a figure corresponding to this plane right here. The value of epsilon shown in this figure is the difference between the actual value of y and the expected value of y or E of y. When x1 takes a certain value x1 star and x2 takes a certain value x2 star in the regression analysis.

Sometimes people use the term response variable instead of the term dependent variable for y. In such a case, in the multiple linear regression equation, we will generate a plane that is known as the response surface. These assumptions seem to be more demanding than what we had for simple linear regression.

Yes, independence across all set of values x1, x2 and so on up to xp, in a sense, these could be more demanding, but they are satisfied in many cases as we will see. And of course, like what we did earlier, we will test for these assumptions before we make use of the multiple linear regression model in decision making.

Next, we will see how to conduct a significance test for multiple linear regression relationship. The significant test that we used in simple linear regression was a t test and an f test. In simple

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linear regression, both of these tests ended up giving us the same conclusion. That is, if the null hypothesis is rejected, we conclude that beta 1 is not equal to 0.

Establishing the significance of a linear relationship between that dependent variable y and the single independent variable x that we had. In multiple linear regression, the t test and the f test have different purposes. What are they used for? The F test in the multiple linear regression is used to determine whether there is a significant relationship between the dependent variable y and the entire set of independent variables x1, x2 and so on up to xp.

Therefore, we refer to this test as a test for overall significance. If the F test shows an overall significance, then the t test is used to determine whether each of these independent variables are significant or not. In this case, we do a separate t test for each one of these independent variables in the model.

And therefore, we refer to these t tests as the test for individual significance. Yes, that is interesting. I am curious to know the details. Yes, let us see the details of the F test and the t test.

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Test of significance using F-test: Introduction

Let us now look at the details of the F test. As we said in multiple linear regression, the F test is used to determine whether there is a significant relationship between the dependent variable y and the entire set of the independent variables x1, x2 and so on up to xp. For this reason, we refer to this test as the test for overall significance.

Recall that the multiple linear regression model is defined as y is equal to beta 0 plus beta 1 times x 1 plus beta 2 times x 2 and so on up to beta p times x p plus epsilon. The hypothesis for the F test all the parameters corresponding to the independent variables namely beta 1, beta 2 and so on up to beta p.

The null hypothesis is that H 0 is given by beta 1 is equal to beta 2 is equal to beta 3 and so on up to beta p is equal to 0. Which means that the alternative hypothesis A is given by the opposite of this statement. Yes. Which is that one or more of these parameters, beta one, beta two, and so on up to beta P is not equal to zero. If the null hypothesis H 0 is rejected, the test indicates that there is significant statistical evidence to conclude that one or more of these parameters beta 1, beta 2 and so on up to beta p is not equal to 0. This implies that the overall relationship between y and a set of these independent variables x 1, x 2 and so on up to x p is indeed significant. However, if H 0 cannot be rejected, then we do not have sufficient evidence to conclude that a significant linear relationship is present. Before describing the steps of the F test, we should review the concept of mean squares. In statistics, we use the idea of mean squares quite often. Recall that the mean square is the sum of squares divided by its corresponding degrees of freedom.

We saw this concept earlier when we did simple linear regression. If you recall, we computed the mean square regression MSR and the mean square error MSE. Yes, mean square regression was sum of squares regression divided by 1. Yes, and do you remember why it was divided by 1? Because there was only one independent variable in simple linear regression.

Yes, so the degrees of freedom used for computing MSR was 1 and therefore, SSR was equal to MSR in that case. Now we will talk about how we extend this concept for multiple linear

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regression. In multiple linear regression case, the total sum of squares SST has n minus 1 degrees of freedom. The sum of squares due to regression SSR has p degrees of freedom.

Is it because there are p independent variables used in the regression model? Yes, exactly. So, m s r, the mean square regression, will be equal to the sum of squares regression SSR divided by p. Now, do you remember how we computed the mean square error for simple linear regression? It was the sum of squares error divided by its degrees of freedom.

Yes, and what was the degrees of freedom? Was it n minus 2? Yes, indeed. It was n minus 2. So in multiple linear regression case, the sum of squares due to error, SSE, has n minus p minus 1 degrees of freedom. If p is equal to 1, as in the case of simple linear regression, it comes to n minus 2. Yes, it does. So our expression for multiple linear regression is a generalization of what we did in the simple linear regression case. Therefore, the mean square due to regression MSR is given by the sum of squares due to regression SSR divided by p, where p is a number of independent variables. And the mean square due to error MSE is given by the sum of squares due to error SSE divided by n minus p minus 1.

Okay, so MSR for multiple linear regression is equal to SSR divided by p. Yes, that is true. And MSE is equal to SSE divided by minus 1. Yes, that's correct. And we saw earlier with simple linear regression that MSE provides an unbiased estimate of the sigma square, the variance of the error term epsilon used in the regression model.

If the null hypothesis is true, that is H naught given by beta 1 equal to beta 2 equal to and so on up to beta p equal to 0, if that is indeed true, then in that case MSR also provides another unbiased estimate of sigma square. And therefore, in this case, the ratio of MSR over MSE should be close to 1. However, if the null hypothesis H naught is false, then the mean square regression MSR will overestimate sigma square and therefore, the value of MSR over MSE will become larger. What do we mean by large? How large is large? That is a good point. In order to determine how large a value of MSR over MSE should it be before we reject the null hypothesis? We will make use of the fact that if the null hypothesis is true and the assumptions of multiple linear regression are valid, then the sampling distribution of MSR divided by MSE will be an F random variable.

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An F random variable with p degrees of freedom in the numerator and n minus p minus 1 degrees of freedom in the denominator. We use this fact to compute the p value to determine how large is actually too large and then we decide on the hypothesis. So in other words, in the F test for overall significance, we have the null hypothesis to be equal to beta 1 equal to beta 2 and so on up to beta p equal to 0.

The alternate hypothesis is that one or more of these parameters is not equal to 0. This gives us a test statistic that is going to be an F random variable, which is the ratio of MSR divided by MSE, where MSR and MSE are defined by SSR divided by P and SSE divided by n minus p minus 1 respectively. The rejection rule for the hypothesis will then be to reject the null hypothesis, and if the p value is less than or equal to the significance level alpha.

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Test of significance using F-test & T-test

Let us now apply the F test on the Hanumantha's data set. Let us look at what the F test tells us for this example. If you recall, Hanumantha was interested in the monthly credit card spent as a function of annual income and household size. We have two independent variables, annual income and household size.

Therefore, the null hypothesis for the F test is written as H naught given by beta 1 equal to beta 2 equal to 0 and the alternative hypothesis is given by beta 1 and or beta 2 is not equal to 0. Here, we see the output of multiple linear regression model in our spreadsheet. If you look at the analysis of variance table as part of the output, you see that the mean square error is given by 437098 and the mean square regression is given by 45734426.

We obtain the F statistic as the ratio of the mean square regression over the mean square error. And what do we get? We get that to be equal to 104. 63. Yes, and for a significance level alpha of 0. 05, we get that the p value is close to 0. This allows us to reject the null hypothesis H naught that beta 1 equals to beta 2 is equal to 0.

Yes, and therefore, we can conclude that there is a significant relationship between the monthly spend y and, the two independent variables, annual income x 1 and the household size x 2. Now, Tejas and Ashwini, if you recall, we had mentioned earlier that the mean square error MSE provides an unbiased estimate of sigma square, the variance of the error term epsilon.

We will see from the ANOVA table output that the mean square error estimate is given by 437098. The square root of the mean square error is the estimate of the standard deviation of the error term. So, what is the square root? It is 661.13. Yes, and we also saw earlier that the standard deviation is called as the standard error estimate and is denoted by S.

Therefore, we have s to be equal to the square root of the mean square error 437098 to be equal to 661. 13. Note that this is the same value of the standard error that is listed just below the adjusted R square value in the regression output. The general analysis of variance or ANOVA as we call it provides a table with the results of the F test for multiple linear regression. This table has a definite structure. We see that the value of the test statistic, the F random variable

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appears on the right hand side of this table along with the significance F which corresponds to the p value of this test.

The degrees of freedom of the F statistic for the numerator and the denominator are given on the left hand side. For the numerator, this degree of freedom is equal to p corresponding to the mean square regression or MSR. For the denominator, this degree of freedom is n minus p minus 1 corresponding to the mean square error or MSE.

The mean square regression MSR is computed as a sum of squares regression SSR divided by p, where p is the number of independent variables. We see that SSR in this case is equal to 91468852, and that divided by two gives us a value of MSR that is equal to 45734426. The main square error MSE is computed as the summer squares error SSE divided by n minus p minus one.

We see that SSE in this case is equal to 22729071 and that divided by 52 gives us MSE to be equal to 437098. Although the mechanics of the F test are similar to what we did in simple linear regression, its implications are different for multiple linear regression. Yes, that is true. We shall see the details of the t test on each of the independent variables.

In multiple linear regression, if we conduct the F test and it shows that the multiple linear relationship is significant, then a t test can be performed to determine the significance of each of the individual parameters. What do we do in the t test of individual significance? For any parameter beta i, we write the null hypothesis H naught as beta i is equal to 0 and the alternative hypothesis H a is written as beta i is not equal to 0.

The test statistic will be a t random variable with n minus p minus 1 degrees of freedom and as we saw in the case of simple linear regression. We will divide b i divided by the standard error in the estimate of b i to compute the test statistic. We will compute this test statistic which would be the value of a t random variable and we will use this value to then compute the corresponding p value.

If the p value is less than or equal to alpha, we reject the null hypothesis. and we claim that there is enough evidence in the sample data that beta i is not equal to 0. This will imply that

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this independent variable x i has a significant role in explaining the dependent variable y i. We do this for each parameter beta i. Yes, we do this test for each parameter beta i.

Let us now conduct the t test for Hanumantha's theta set. If you look at the output from the regression analysis, we will see the details of the t test calculations. Let us start by first looking at the independent variable x 1 corresponding to annual income.

The null hypothesis of the t test would be that H naught is given by beta 1 is equal to 0. The alternative hypothesis then would be given by beta 1 not equal to 0. The regression output then gives us an estimate of b 1 from the sample data to be equal to 71. 17.

The standard error s of b 1 is given to be equal to 6. 92. Then the test statistic corresponding to this parameter beta 1 is given by 71. 17 divided by 6. 92. And what does that come out to be? That comes out 10.28. Yes, this statistic is a t random variable with n minus p minus 1 degrees of freedom, which is 52 degrees of freedom in our case.

And that gives us a p value that is fairly close to 0. Therefore, we can reject the null hypothesis that beta 1 is equal to 0. And conclude that the association between annual income and monthly spend is indeed significant.

Next, let us look at the second independent variable X2 corresponding to the household size. The null hypothesis of this t test would be that H naught is given by beta 2 equal to 0. The alternate hypothesis then would be given by beta 2 not equal to 0. The regression output gives us an estimate of b 2 from the sample data to be equal to 339. 92. The standard error s of b 2 is given to be equal to 54. 05. Then the test statistic corresponding to this parameter beta 2 is given by 339. 92 divided by 54. 05. And what does that come out to be? That comes out to be equal to 6. 29. And this test statistic will be a t random variable with n minus p minus 1 degrees of freedom 2, which is 52 in our case. And this gives us a p value that is also fairly close to 0.

Therefore, we can reject the null hypothesis that beta 2 is equal to 0. So, putting the results of these two three tests together, we can conclude that the association between household size and monthly spend is significant and the association between annual income and monthly spend is also significant.

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Multicollinearity

Now we have been using the term independent variables in regression. And when we use that term, the independent variable in regression analysis, we use it to refer to any variable being used to predict or explain the value of the dependent variable. Now, just because we use the term independent variable, it does not necessarily mean that the independent variables themselves are independent in the statistical sense.

In fact, the opposite is often true. Most independent variables that we use in multiple linear regression are correlated with each other to some degree. Was that the case for Hanumantha's data set? Yes. For example, in Hanumantha's data set, we have two independent variables, x1 corresponding to annual income and x2 corresponding to the household size.

And you mean to ask if the annual income x1 and the household size x2 were linearly correlated? Yes, our annual income x1 and household size x2 linearly correlated. Well, how would we determine that? Perhaps we could create a scatter plot of x2 versus x1. Yes, we could do that. We could also treat the annual income as a dependent variable y and consider the household size as an independent variable x and determine if these two variables themselves are related by trying to find a simple linear regression equation linking annual income y to the household size x.

We can also compute their correlation coefficient. Yes, sure. You can compute the sample correlation coefficient between monthly spend and annual income and household size. In fact, I have this correlation with me. We get a sample correlation coefficient of 0. 32.

How do you get it? Okay. Well, let's see the details then. In order to compute the correlation coefficient between household size and annual income, we can use analysis tool pack again. And just like we found descriptive statistics and regression outputs, analysis tool pack also has an option to compute sample correlations. We choose that option from the menu and then for the input data, we can choose the data in columns B, C and D corresponding to annual income, household size and monthly spend and obtain the output.

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We obtain a table as shown here that has the correlation coefficients between each pair. The correlation between monthly spend and household size, the correlation between monthly spend and annual income as well as the correlation between annual income and the household size. From this table, we can see that the sample correlation coefficient between the household size and the annual income is 0. 32. This indicates that there is some degree of linear association. Yes Tejas, there is some degree of linear association between these two independent variables, annual income and household size that we used in the regression.

So now what happened? In multiple linear regression, multicollinearity refers to the correlation between independent variables that we use in the regression model as we just saw. Multicollinearity can create problems in multiple linear regression. To understand the potential problems that multicollinearity can create, let us consider a hypothetical scenario, a variation of Hanumantha's data set. Let us say that instead of using household size as the independent variable, Hanumantha had a column corresponding to monthly income.

You mean annual income as one column and monthly income as a second column? Yes, exactly. You can see here an illustration of what I am implying. Then I have added a new column which corresponds to monthly income. Then Column C, column D and column E have the original data corresponding to annual income, household size and monthly spent.

And now let us say for our multiple linear regression, we had chosen the independent variable x 1 to be equal to monthly income and the independent variable x 2 to be equal to the annual income. Clearly, we expect that the annual income x 2 and the monthly income x1 are likely to be highly correlated independent variable.

Can we find their correlations? Yes, indeed, using analysis tool pack, using what you just showed us. Yes, so we choose the correlation option, select columns B, C, D, and E, and compute the correlations. You see that the monthly income and the annual income have a high correlation of 0. 83. We would expect that, of course. Yes, and notice that the other pairwise correlations are the same as we got earlier, as we expect. Yes, that is true.

Now, let us try to find the regression coefficients b 0, b 1 and b 2 using x1 to be monthly income and x2 to be annual income. What do we get? Let's go back to the analysis toolpath and see

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what we get. Okay, we choose the regression option. Yes, and column E with the monthly spend as your dependent variable y and columns B and C as your independent variables x1 and x2. What is your output? The R square is 0. 66, not very good, similar to what we had with only annual income.

Yes, and what else? F test looks okay. P value is small, indicating linear relationship between Y, X1 and X2. Yes, and what about the T test and the P values for X1 and X2? The T value corresponding to monthly income X1 is 0. 18 and the T value corresponding to n1 income X2 is close to 0.

Yes, and what do we infer from these p values? It is suggesting that the null hypothesis corresponding to x1, that is beta 1, equals to 0, cannot be rejected. Yes, and how can we explain that? Which means that beta 1 must be 0. Yes, and if that is indeed the case, are we saying that monthly income has no impact on monthly spend?

Surely that need not be true. So, how do you interpret that output? Well, what it is really trying to say is that with annual income as x 2 that is already present in the model, it does not add any significant value to have monthly income as x 1 also in the model. This choice of x 1 does not make any significant contribution for determining the value of the monthly spend y.

Yes, this interpretation makes more sense in our example. Yes, and if we see the p value corresponding to the annual income x2, we see that it has a p value that is close to 0. This suggests that the null hypothesis corresponding to beta 2 equal to 0 can be rejected, indicating that there is a significant association between annual income x2 and the monthly spend y.

So, knowing that the monthly spend and the annual income have a significant association, adding monthly income into the model as another independent variable does not really add much. So, to summarize, what is the difficulty caused by multicollinearity? We see that the issues related to multicollinearity are observed when we do the individual t test.

We see here that in Hanumantha's case, that the t stress are indicating that some independent variables like the monthly income in this case that we believe should have a strong effect on the dependent variable monthly spend y in this case, need to be removed from the model. In

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fact, multicollinearity may suggest that none of the individual parameters are significantly different from 0.

That is all beta i could be equal to 0 in some cases. when an F test on the other hand on the overall multiple regression equation might indicate a significant relationship. Of course, this problem is not there when there is little or no correlation between the independent variables. How do we know how much correlation is significant?

That is a good question. The correlation between monthly income and annual income was 0. 8. What was the correlation between household size and annual income? It was 0. 324 and that did not cause multicollinearity problems. Yes, and so as we said, we do not expect that our independent variables x1 and x2 to be statistically independent.

However, if they have high correlation, we could see problems due to multicollinearity as we just saw. As a rule of thumb, if a sample correlation coefficient of greater than plus 0. 7 or less than minus 0. 7 is obtained between two independent variables, then it is a warning that there could be potential problems related to multicollinearity.

That is why we saw multicollinearity issues with monthly income and annual income. Yes, but it is a rule of thumb not set in stone. And in statistics, there are several tests that are used to determine whether the multicollinearity that you see is high enough to cause problems. But we will not go through them here.

But you should remember that if multicollinearity is detected, like we did in this modified case of Hanumantha's data set, then every attempt should be made to avoid including those independent variables that are highly correlated into the model. So we go back to our earlier model with annual income and household size.

Yes, indeed. And that is a good model. However, sometimes in practice, it is hard to adhere to this policy strictly. When independent variables are highly correlated, It may not be possible to determine the separate effect of any particular independent variable on the dependent variable. Usually, in most cases, multicollinearity does not affect the way in which regression analysis is performed or interpreted.

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However, when the multicollinearity is severe, that is, when two or more independent variables are highly correlated to another, we can have difficulty in interpreting the results of the test of the individual parameters. Sometimes severe multicollinearity can result in the least squares method yielding regression coefficients that could have the wrong sign.

What do you mean? Well, actually Ashwini, you can see that happening in our case with multicollinearity. We only looked at the p values. Look at the corresponding estimates of b 1 and b 2. What are they? The coefficient b1 for monthly income is minus 202. 85 and the coefficient b2 corresponding to annual income is 102. 74. Yes, and we know that intuitively as either of them increases, I mean as either monthly income x1 or annual income x2 increases, the dependent variable y, monthly spend should also increase. So, we expect both B1 and B2 to have positive sign, but B1 is negative. I see what you mean. Yes, and that is what I was referring to. Yes, that is what I was referring to. And so now, that impacts the interpretation of both B1 and B2. And we end up having little faith in the individual coefficients, if multicollinearity is present to this high of a degree. Therefore, care must be taken to not include variables that have high linear correlation as part of the independent variables used in multiple linear regression.

We were good with what we did first. We are back to the model with x1 as annual income and x2 as household size. Yes, but it is important to know the issues that we could have faced with the wrong choice of the independent variables. Let us now talk about the residual analysis. And how looking at the residuals, we can validate our model assumptions.

As we said earlier, the residual for the i th observation is the difference between the observed value of the dependent variable y i and the predicted value of the dependent variable y i hat. That is, the residual is the error resulting from using the estimated regression equation to predict the value of the dependent variable.

In the Excel output, we can see the residuals for Hanumantha's data set. For each of the 55 observations, we find the value predicted for monthly spent from the estimated regression equation and the residual is then obtained by comparing this predicted value with the actual value in the data set. If you recall, we had made some assumptions regarding this error term

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epsilon in the regression model, y is equal to beta 0 plus beta 1 times x 1 plus beta 2 times x 2 and so on up to beta p times x p plus epsilon.

Let us remind ourselves of some of those assumptions. What were those assumptions? Do you remember? We had sent a error absolute as random variable. Yes, and the expected value of epsilon is equal to 0. And as a result, the expected value of Y, E of Y will be equal to beta zero, plus beta one times x one plus beta two times X two, and so on up to beta p times x p.

What else? We had said that the error epsilon is a random variable. We assumed that the variance of epsilon, the error term denoted by sigma square is the same for all values of x 1, x 2 and so on up to x p. Yes, and so in other words, the variance of y about the regression line equals sigma square and it is the same for all values of x 1, x 2 and so on up to x p. This property is also known as homoscedasticity.

Any other assumptions? We assume that the error term epsilon are independent of the xi. And since the value of epsilon for any particular set of values of x1, x2 and so on up to xp is not related to the value of epsilon for any other set of values of x1, x2 and so on up to xp, Then, the value of y for a particular set of values of x 1, x 2 and so on up to x p is not related to the value of y for another set of values of x 1, x 2 and so on up to x p.

There was another assumption, do you remember? Yes, I think it was that the error term epsilon is normally distributed random variable for all set of values of x1, x2 and so on up to xp. Yes and this assumption along with our earlier assumptions imply that we are assuming that the error term epsilon is normally distributed with mean 0 and variance equal to sigma square and also because y is a linear function of epsilon going back to the properties of linear combinations of random variables, we can say that our dependent variable Y is also normally distributed random variable for all values of X.

These assumptions were vital to being able to justify and conduct the t test and the f test that we did recently to check whether the relationship between the X i's and the Y i are statistically significant. If the assumptions about the error term epsilon do not hold, then the hypothesis test and the significant test of the regression relationship may not be valid.

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The residuals provide the best information about these error terms epsilon. And hence, an analysis of these residuals helps us to determine whether the assumptions related to epsilon are appropriate. Now, like we did with simple linear regression, Much of the residual analysis is going to be based on examining some of these graphical plots.

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MLR - Residual Analysis

Here, we will look at three types of plots. The graphical plot of the residuals against each of these independent variables x i, a plot of the residuals against the predicted value of the dependent variable y, and a standardized residual plot against the value of the dependent variable y. Let us look at each one of them.

In multiple linear regression also, the residual plot against the independent variable x i is a graph in which the values of the independent variable x i are represented on the horizontal axis and the corresponding residual values are represented on the vertical axis. A point is plotted for each residual in this manner.

Will we get a plot for each independent variable x i? Yes, that is true. We can discuss the details using Hanumantha's dataset and the residual plots for annual income x1 and household size as x2. Ashwini and Tejas, when we got the outputs from the analysis tool pack, I hope you selected these options to get these plots.

Yes, I got them. Tejas, do you have them? No, I don't seem to have them. What do I do? Well, you can reselect the data, the Y values and the XI values and this time, remember to check the options so that you get the residual plots. Okay. Just a minute. Sorry. No problem. We will wait for you. Okay. Now I have regression outputs and the residual plots.

We can see that the analysis tool pack output has provided us a plot of the residuals against the values of the independent variable x1, which is the annual income. And it has provided a plot of the residuals against the second independent variable x2, which is household size. Before interpreting the results of these residual plots for Hanumantha's data set, let us recall what we are concerned about.

Are we looking for a band like before? Yes, by and large, we are usually interested in three patterns that are summarized in this figure. If you recall, we made an assumption that the variance of epsilon is the same for all values of x, and that the assumed regression model is an adequate representation of the relationship between the variables.

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If that is the case, then then the residual plot should give an overall impression of a horizontal band of points, such as the one in panel A. However, if the variance of epsilon is not the same for all values of x, in particular, if the variability about the regression line is greater for larger values of x, then a funnel pattern such as the one showed in panel B would be observed.

In this case, the assumption of constant variance, or homoscedasticity, is being violated. Another possible residual plot is the one shown in panel C, which is roughly a U shape. Here, we can conclude that the assumed regression model is not an adequate representation of the relationship between the variables.

So, looking at the plots, what can we say about the residual plot for Hanumantha's dataset? What does the plot of the residuals against x1, which is annual income look like? It looks more like a band rather than a funnel or a U shape. Yes. And what about the plot of residuals against x2, the household size?

This looks even more like a band. Yes, I agree. So, can we say that our assumptions regarding epsilon seem to hold? I would imagine that to be the case. Now, the residual plot is one technique used to assess the validity of the assumptions of a regression model. As we saw earlier, we can also plot residuals against the predicted values y hat instead of the individual x i values.

This is particularly useful if you have several independent variables, not just two as in our example right here.

We saw this with simple linear regression too. We can also have a residual plot where the predicted value of the dependent variable y hat is plotted on the horizontal axis and the residual values are on the vertical axis. As before, a point is plotted for each residual point. Note that the analysis tool pack does not automatically give us this residual plot.

But this residual plot can be easily obtained using the outputs provided by the analysis tool pack. And what do we see? In my opinion, at least for Hanumantha's dataset, this does not present any cause for concern. The pattern for the residual plot is roughly a mix of the random plots that we saw for the individual dependent variables Xi's.

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It has roughly the same random pattern. Agree? Yes, I think so. Note that in multiple linear regression, this residual plot with y hat is more widely used because of the presence of more than one independent variable. In fact, usually we look at this plot first. And then, if needed, we look at the plots of the residuals against the individual independent variables Xi.

The Analysis Tool pack output also provides us the standardized versions of the residuals. And as we have seen earlier, a random variable can be standardized by subtracting its mean and dividing this result by its standard deviation. Now, since we have obtained these residuals using the least squares method, the mean of these residuals is equal to zero.

Therefore, simply by dividing the residual by its standard deviation gives us the standardized residuals. We can then compute the standardized residuals and plot them against the independent variables x. For multiple linear regression, the exact equation to compute the standardized residuals for an observation I is hard to do by hand using a closed form expression.

The main challenge is in computing the standard deviation of these residuals. However, statistical software including the analysis tool pack in Excel does provide the standardized residuals as part of the output. Yes, of course. Excel is our best friend. Sure. When Excel gave us the residuals, it also computed the standardized residuals.

Note that, for Hanumantha's example, where we had the monthly spend as the dependent variable y, and the annual income as the first independent variable x1, and the household size as the second independent variable x2, the predicted values used in the computation of these residuals were obtained by using the equation having B0 equal to 355. 7, B1 equal to 71. 17, and B2 to be equal to 339. 92. This implies that the predicted monthly spend y hat is given by 355. 7 plus 71. 17 times the annual income in lakhs x1 plus 339. 92 times the household size x2. Subtract this from the actual observation and we get the residual.

Yes Tejas and the plot of the standardized residuals provide us insights about the assumption that the error term epsilon has a normal distribution. If this distribution is normal then the distribution of the standardized residuals should come from a standard normal probability distribution. Thus, when looking at the standardized residual plot, we should expect to see roughly 95 percent of the standardized residuals between minus 2 and plus 2.

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Yes, 95 percent of the residuals should lie within plus and minus 2. Exactly. And so if you look at the standardized residual plots for Hanumantha's data set, we see that is indeed the case. This is similar to what we did for the simple linear regression. Excel makes residual analysis easy. Yes, Ashwini, that is correct.

Manual computation can be cumbersome, especially if the data set is large. However, most statistical software have inbuilt capabilities to compute these outputs. So that our focus then is on the interpretation of the outputs from regression. Now that we have seen how residual analysis can be used to determine whether violations of the assumptions about the regression model are happening.

We will now discuss how residual analysis can be used to identify observations that can be classified as outliers. Some steps need to be taken when such observations are seen.

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MLR - Outliers and Influential Observations

An outlier is a data point that does not fit the trend shown by the remainder of the data. As a first check, scattered plots can help us identify potential outliers. We saw that during our discussion on simple linear regression, how standardized residuals can also be used to identify outliers.

Standardized residuals can also be used to identify outliers in multiple linear regression. If an observation deviates greatly from the pattern of the rest of the data, the corresponding standardized value will also have a large absolute value. For normally distributed errors, a standardized residual should be in the range of minus 2 to plus 2, 95 percent of the time.

Therefore, data points corresponding to cases where the standardized residuals lie outside the value of minus 2 or plus 2 could be considered as outliers in the data set. In Hanumantha's data set, we do see a few observations with standardized residuals that have absolute values that are greater than 2.

What action do we take when we see outliers? Outliers may represent erroneous data and if that is the case, they should be corrected. Alternatively, they may represent a violation of the model assumptions. And if that is the case, another model should be considered. Finally, they may just be unusual values that are occurring by chance and could be very well retained.

Let us now talk a little bit about detecting influential observations. Sometimes one or more observations exert a strong influence on the results obtained for multiple linear regression. If such an influential observation is drawn from the data set, then the slope of the estimated regression equation could actually change and the parameters of the regression model could change as well.

How do you detect influential observations? Yes, an influential observation. may be an outlier or an observation with a y value that deviates substantially from the trend. Alternatively, it may also correspond to an extreme value of the independent variable x i that is far from its mean or it may be caused by a combination of both of these phenomenon.

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Now, since each influential observation may have a dramatic effect on the estimated regression equation, they must be examined carefully. What do we do with them? We should first check to ensure that no error was made in collecting or recording the data. If an error was made in recording the data, it can be corrected and the new estimated regression equation should be developed.

If the observation is indeed valid, then such observations could contribute to a better understanding of an appropriate multiple linear regression model, and that can lead to a better estimated regression equation. In the presence of an influential observation, especially when it is valid, it is recommended to try and obtain data on the intermediate values of x i

to better understand the relationship between Xi and Y. In some advanced text, you will find more detailed measures to identify outliers and influential observations. Some statistical packages also provide options to compute such measures as well. However, we will not discuss these techniques as part of this module.

The procedures for identifying outliers and influential observations provide warnings about the potential effects of some observations and their effect on the results of the regression equation. Each outlier and influential observation requires a careful examination. If data errors are found, they can be corrected and the regression analysis can be repeated.

In general, outliers and influential observations should not be removed from the dataset unless there is clear evidence that they are not based on the elements of the population being studied and they should not have been included in the original dataset in the first place.

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MLR – Recap

We have covered a lot of important concepts related to multiple linear regression. As we covered each concept, we applied these concepts to Hanumantha's dataset. Let us review what we did before we apply these concepts to a few other examples. In multiple linear regression, the main difference from simple linear regression is that we are trying to establish a relationship between a single dependent variable y and several independent variables x i ranging from 1 to p.

The linear regression model is then y is equal to beta 0 plus beta 1 times x 1 plus beta 2 times x 2 and so on up to beta p times x p plus epsilon. The idea is to use the data as a training set and derive estimates for the parameters beta 0, beta 1, and so on up to beta p. We denote these estimates as b 0, b 1, and so on up to b p to distinguish them from the original parameters in the model.

When we get these estimates, it gives us what is called as the estimated linear regression equation. Y hat is equal to b 0 plus b 1 times x 1 plus b 2 times x 2 and so on up to b p times x p. We use the ordinary least square method to identify these estimates b 0, b 1, b 2 and so on up to b p. The ordinary least square method tries to minimize the sum of y i minus y i hat square, which we also called as the sum of squares error or SSE.

Then to determine how well the estimated regression equation explains the data, we computed the coefficient of determination or R square. To compute this R square, we computed the total sum of squares SST, which was the sum of yi minus y bar square. And then we split this SST into two parts. We write SST as the sum of SSE and SSR.

Yes, where SSE is given by the summation of yi minus y hat square. And SSE is given by SSR is given by the summation of y i hat minus y bar square. Using these sum of squares SST , SSR and SSE, the multiple coefficient of determination was then obtained. It was similar to the coefficient of determination in simple linear regression.

Do you remember what it was? We said that R square is equal to SSR by SST. Yes, and we saw that this can take a value between 0 and 1. And higher that value, better is the ability of

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the regression model to explain the variation in the observed data. We also saw an adjusted R square measure. Do you remember what that was?

Yes, it accounted for the number of independent variables in the multiple linear regression model. Yes, it is usually slightly lower than the simple R square. Then what did we do? Test of significance. The F test for overall significance and the set of T test for significance of each independent variable.

Very good. Yes. We did them to determine the significance of the linear relationship that is being hypothesized. In order to do that, we had to make some assumptions regarding the error term epsilon in the regression model. These assumptions were critical in the hypothesis test that we did. In the F test for overall significance, the null hypothesis of the F test is written as H not is given by beta 1 equal to beta 2 equal to beta 3 and so on up to beta p equal to 0. And the alternative hypothesis was given by not all of these betas are equal to 0. This F test then required an estimate of the variance epsilon which we denoted by sigma square.

We use the data available to obtain the ratio of MSR over MSE, which we argue to be an F random variable. If the P value corresponding to this F random variable was small, then we were able to reject the null hypothesis and say that the coefficients corresponding to at least one or more of these independent variables are not equal to zero and the overall relationship between y and the xi's are indeed significant. Then we could do the individual t test. What did we do for the t test? For the t test, we computed the t statistic corresponding to each independent variable. Yes, this t statistic had a degree of freedom of m minus p minus 1.

Using this t statistic, we computed the p value and we reject the null hypothesis if the p value is small enough. Rejecting the null hypothesis allowed us to claim that beta i is not equal to zero and therefore the linear relationship between x i and y i is significant. Then we looked at the idea of multicollinearity.

What was it basically saying? It is said that the independent variables Xi could have, statistically speaking, linear correlation. Yes, and if that is high, then interpreting the outputs of multiple linear regression analysis will be hard. And how do we detect multicollinearity? The p value of independent variables might be more than 0. 05. Yes, and any other signs or

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indicators? The signs of Bi might be opposite to what we expect. Yes, well done both of you. Some correlation will always be there and that is okay, but if it is severe, we may want to exclude some variables from the model.

Finally, we looked at the analysis of the residuals to determine whether the assumptions that we made for the test of significance on the error epsilon were violated or not. We looked at the plots of the residuals against the independent variables xi and the plot of the residuals against the dependent variable y. We also looked at a plot of the standardized value of the residuals to see if there were any outliers.

So, that was the theory.

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MLR - Application Recap

And what did we find when we did all of this for Hanumantha's data set? We assumed monthly spent to be the dependent variable y and the independent variables to be annual income as x1 and household size as x2. We found that the estimate of the regression parameters b0, b1 and b2 were 355.7, 71.17 and 339.91 respectively. Yes, and therefore, the estimated regression equation was given by 355. 7 plus 71. 17 times x1 plus 339. 92 times x2.

The model had an R square of 0.80 far better than simple linear regression with annual income as independent variable. Yes, simple linear regression gave us an R square of 0. 65, and this improved to an R square of 0. 80, saying that including the two independent variables that we modeled were able to explain close to 80 percent of the variation observed in the monthly spend Y.

What else did we find? We saw that the adjusted R squared is 0. 79, close to R squared of 0. 80. Yes. And what else? We found that the standard error S is given by 661. 1. Yes, that is correct. And what did the ANOVA table tell us? The F statistic computed as the ratio of MSR over MSE was 104. 6. Yes, and that implied a pretty low p value which meant that we could reject the null hypothesis and conclude that the overall linear model has some significance. And what did the t test tell us? To compute the test statistic for each t test, we divided b i by the standard error.

Yes, and for both of the t tests, our p values were close to zero, which means that we can reject the null hypothesis that says that those beta i's are equal to zero, implying that the linear relationship between the y and the xi's is indeed statistically significant. And did we find any multicollinearity? Not with these choices of x1 and x2. Yes, but when we added monthly income as another independent variable, we saw that theory in action. Good. And what did we find from our residual analysis?

We computed the standardized version of the residuals. Yes, and we found that most of these residuals are within plus 2 and minus 2. And there were a few values that had standardized residuals that were slightly above 2 in an absolute sense. And what about the plots of these residuals? Both did not indicate any problematic patterns.

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Yes, and although they were not perfect bands, they did not show to have any funnel or curvilinear shape either. Excellent. Now we are ready to look at another example. Are we going to revisit one of the examples we did with simple linear regression? Yes, indeed, because that will allow us to compare.

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MLR - Applications with Examples – I

We are visiting Manjula Nayak. If you recall, Manjula Nayak, who works for 99acres, had a data set from a household survey done in Bengaluru with around 500 observations. It has various types of information, including the number of members of the household, where they reside, their monthly income, monthly payments and so on.

We will look at these fields in more detail next. Manjula is working with banks to help customers get home loans and she wants to know what factors influence a household's debt. She wants to know if the household income is a significant determinant of the household debt. We did a few types of simple linear regression for her.

Do you remember what was our dependent variable? Debt was our dependent variable y. That is right and what was our independent variable? We did one with first income and the other with monthly payment. Yes, now we can do a multiple linear regression with both of those as x1 and x2. Let us now try to construct a linear regression of the debt Y as a function of the first income X1 and monthly payment X2.

Here, I have the data in the spreadsheet with debt in column K and the first income in column H and monthly payment in column I. Let us conduct the regression analysis using the analysis tool pack and interpret the outputs. In the analysis tool pack, we select the y variable to be the household debt data in column K and the x variables to be the first income in column H and the monthly payment in column I.

As we have done before, in addition to obtaining the output, let us also compute the residuals, the standardized residuals and the plots of the residuals. I will wait till you collect all of that.

Okay. Now that we have the outputs, what do we look at first? Let us look at R square. We get a value of 0. 45 from tool pack. Yes, and did you know what it was for the simple linear regression with only first income as the independent variable? It was 0. 31 with only first income as the independent variable.

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Yes. And what was the R square value when we had monthly payments as our only independent variable? It was 0. 37 in the case where only monthly payment was the independent variable. Yes, so this implies that our multiple linear regression model is able to do better. It is able to explain 45 percent of the variations in the household debt as a function of the first income of the household and the monthly payment.

Clearly, debt has a relationship with first income and monthly payment, but it might also depend on other factors it appears, because our R square is only 0. 45. Now, what does the output tell us about the adjusted R square value? The adjusted R square is pretty close to R square. Yes, it is around 0. 448, almost 0. 45. And that is because we have a large value of n here of about 500. Now, let us look at the standard error. We see that the standard error here is 25644. How does it compare to the standard error that we had for the two simple linear regressions we did earlier? For the earlier simple linear regression, we had standard errors that were more.

Yes. We had a standard error of 2 8 6 1 9 when we had only first income and a standard error of 2 7 5 1 8 when we had only monthly payments. Having the two independent variables has reduced the standard error and increase the R squared. Next, let us examine the ANOVA output. The table shows that the sum of squares regression and the sum of squares error, they're big numbers as we had earlier.

Yes, the SSR is equal to 2.68 10 to the power 11 and SSE is 3.27 times 10 to the power 11. And what is SST? SST is given by 5. 94 times 10 to the power 11. Now notice that this is the same as what we saw with the simple linear regression, right? Yes, and SSR were lower in the earlier cases. Yes, with only one first income as the independent variable, the SSR was 1.6 times 10 to the power 11. And with only monthly payment as the independent variable, The SSR was 2.18 times 10 to the Power 11. Now with two independent variables, first income and monthly payment, the SSR has increased to 2.68 times 10, to the power 11. The two independent variables are able to explain more of the variation in monthly debt.

Yes. And what about MSR and MSE? MSR is equal to 1.34 times 10, to the power 11 and MSE is 6 5 7 6 2 4 0 5 4 with two independent variables. When we had only first income, the MSR was 1.87 times 10 to the power 11 and MSE was 8 1 9 0 7 6 0 9 7. When we had only monthly

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payment as the independent variable, MSR was 2.18 times 10 to the power 11 and MSE was 757269785.

Since we are looking at multiple linear regression with two independent variables, the degree of freedom for the mean square regression is equal to 2 and for MSE, the degree of freedom is 497. Okay, what does the F statistic come out to be? The F statistic is the ratio of MSR over MSE and that comes out to be 203.

Yes, and what does that mean in terms of the p value? It is close to 0 and therefore we conclude this allows us to reject the null hypothesis that beta 1 is equal to 0. Yes, and that implies that there is enough evidence to claim that beta 1 and beta 2 are both not equal to 0. And therefore, there is a significant overall linear relationship between debt, first income and monthly payment.

Next, let us look at the table with the coefficients. What are b 0, b 1 and b 2? B0 is 13081, B1 is 0. 025 and B2 is 2. 09. Yes, and how do you interpret the coefficient B1? For every rupee increase in the first income, monthly payment being held constant, the debt increases by 0. 025, right? And, how do you interpret the coefficient B2? For every rupee increase in the monthly payment, the first income being held constant, the debt increases by 2. 09.

But are these coefficients significant? Could beta 1 or beta 2 be equal to 0? What is the outcome of the hypothesis test with the null hypothesis that beta 1 is equal to 0? We need to look at the row corresponding to first income and its coefficient b1. Yes, and what is the standard error, the test statistic and the p value?

We notice that the standard error is 0. 003. And if we divide the coefficient b1, which is 0. 025 by the standard error of 0. 003, we get. The t statistic which is equal to 8. 74. Yes, and for this t statistic, what is the p value? It is very, very small. And what does that imply? It implies that we can reject the null hypothesis, again confirming that the linear relationship between first income x1 and household debt y is indeed significant.

And what is the outcome of the hypothesis test with the null hypothesis of beta two equal to zero. For that, we look at the next row corresponding to monthly payment and it's coefficient

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B two. Yes. And what is the standard error, the test statistic and the P value? We notice that the standard error is 0.188, and if you divide the coefficient B two, which is 2.09, by the standard error of 0.188, we get. The t statistic with 497 degrees of freedom, we get a value of 11. 1. Yes, and for this t statistic, what is the p value? It is again very, very small. And what does that imply? It implies that we can reject the null hypothesis, again confirming that the linear relationship between monthly payment x2 and the household debt y is indeed significant.

Let us now look at the residuals. We have the residuals as well as the predicted debt given by our estimated regression equation. Notice that the predicted debt is the expected debt for a given value of first income and monthly payment. What can we say about the standardized residuals? Here, we can see that there are a few values of standardized residuals that are less than minus 2 and more than plus 2.

Yes, and by and large, they are within the range. And what does the plot of residuals against x1, which is first income, tell us? The plot of residuals against x1 looks like an oval. And what about the plot of residuals against x2, monthly payment? That is also similar. Yes, they both do not have a perfect band shape, but neither does it have a funnel or a curvilinear shape.

What about the plot of residuals against y hat, the predicted values of debt? Again, this plot is over. Yes, again, nothing to be concerned about. And what about the plot of standardized residuals against y hat, the predicted values?

It reveals the same oval pattern. There are a few values of standardized residuals that seem to lie above and below 2. Yes, we expected that too because we saw them in the table of standardized residuals also. Can we do a multiple linear regression of debt versus 2 variable? Sure, which two variables shall we pick?

Monthly payments and utilities. So you mean leave out the first income but include utilities instead? Yes, let's try that. Okay, let us try that next.

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MLR - Application with Examples - II

For the household survey data that Manjula Nayak did, we developed a multiple linear regression model with debt y as a function of the first income x 1 and monthly payment x 2. The estimated regression equation was 13081. 6 plus 0. 0246 times the first income plus 2. 09 times the monthly payment. The model had an adjusted R square of 0. 448. Both the F test for overall significance and the T test for individual significance showed that the assumptions behind the multiple linear regression model hold. And the linear relationships are significant. The outlier analysis also did not reveal any concerns. Now Ashwini wants us to conduct a multiple linear regression model using debt as a dependent variable and monthly payment and utilities as the independent variables X1 and X2.

Before we get into this, Ashwini, I am curious, what are you hoping for? Ashwini? I don't know really. I thought it would be good to see the process carried out for a different set of independent variables. Okay. And Tejas, any expectations from this new regression model? I am hoping we can compare how the two models perform. We will learn something. Okay. Let's do it and see what we find. Here, I have the data in the spreadsheet. With debt in column K and monthly payment in column I and utilities in column J. Let us conduct the regression analysis using the analysis tool pack and interpret the outputs. In the analysis tool pack, we select Y variable to be the household debt data in column K and the X variables to be the monthly payment in column I and the utilities in column J.

As we have done with our previous models, in addition to obtaining the output, let us also compute the residuals, the standard residuals and the plots of the residuals. Make sure you get the plots too Tejas. Yes, yes, I will not forget this time. Okay, looks like Ashwini has already got the outputs. What shall we look at first?

Let us look at R square. What did you get? We get a value of 0. 67 from ToolPak. Oh, good. That is better than the 0. 45 that we got with first income and monthly payment as the independent variables. This looks promising. Yes, and for the simple linear regressions with only first income as the independent variable, the R square was 0. 31. And for the simple linear regression with only monthly payment as the independent variable, the R square was 0. 37. So

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this implies that our multiple linear regression model is able to do better. It is able to explain 67 percent of the variation in the household debt as a function of monthly payment and utilities.

It appears the variation in debt is better explained through utilities as compared to low first income. I don't see why more utility payment should cause more debt. Well, Tejas, linear regression is not really establishing causality, remember? Only association. So, let us not jump into conclusions. Yes, I should remember, only association, no causality.

Yes, okay. Now, what does the output tell us about the adjusted r square? The adjusted r is pretty close to the r square. It is also around 0. 67, just slightly lower than the R square that we computed, perhaps the effect of a large value of n again. Next, let us look at the standard error. We see that the standard error is 19839, which is lower than the standard error of 25644 that we got with first income and monthly payment as our independent variables.

And how does it compare with what we had for the two simple linear regressions? I think for the earlier simple linear regressions, we had standard errors that were more. We had standard error of 28619 when we had only first income and It looks like we have done better now and we had a standard error of 27518 when we had only monthly payments.

Yes, so having monthly payments and utilities as the two independent variables has reduced the standard error and increased the R squared. Now let us examine the ANOVA output. The table shows that SSR is equal to 3. 99 times 10 to the power 11. And SSE is 1. 96 times 10 to the power 11. The SSR is higher than the value of 2. 67 times 10 to the power 11 that we got for the previous model. The SSE is lower than 3. 27 times 10 to the power 11 that we got for the previous model. And what is the SST? The SST remains at 5. 94 times 10 to the power 11. It should, right? Because the total variation in the debt is independent of the x1 and x2 that we pick for our model.

Yes, the higher SSR in this case is saying that the multiple linear regression with these variable is able to explain more of the variation in debt than the other choices. Yes, and with only first income as x, the SSR was 1. 86 times 10 to the power 11. With only monthly payment as x, The SSR was 2. 18 times 10 to the power 11.

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And with the two independent variables that we had chosen earlier, first income and monthly payment, the SSR increased to 2. 67 times 10 to the power 11. Now with monthly payment and utilities, the SSR is even better, 3. 99 times 10 to the power 11. And what about the MSR and MSE? The MSR here is 1. 99 times 10 to the power 11 and the MSE is 393601978 with two independent variables and since we are looking at a multiple linear regression with two independent variables, the degree of freedom for mean square regression is equal to 2 and for the MSE the degree of freedom is 497.

Ok. What does the F statistic come out to be? The F statistic is the ratio of MSR over MSE and that comes out to be 506. 9. Yes. And what does that mean in terms of the P value? It is close to 0 and therefore, we conclude this allows us to reject the null hypothesis that beta 1 and beta 2 are both equal to 0.

Yes. And that implies that there is enough evidence to claim that beta 1 and beta 2 are both not equal to 0. And therefore, there is a significant overall linear relationship between debt, monthly payment and utilities.

Next, let us look at the table with the coefficient. What are B zero B one and B2, B zero is minus 1 4 0 7 7 1 , B 1 is 1.44 for monthly payment and B2 is 168.91 for utilities? Yes. And how do we interpret the coefficient B1? For every rupe increase in monthly payment, utility is being held constant. The debt increases by 1444. For every rupee increase in monthly payment, utilities being held constant, the debt increases by 1. 44, right? And how do you interpret the coefficient B 2?

For every rupee increase in the utilities, monthly payment being held constant, the debt increases by 168. 91. Then, the next question is, are these coefficients significant? Could either beta 1 or beta 2 be equal to 0? What is the outcome of the hypothesis test with the null hypothesis that beta 1 is equal to 0?

We need to look at the rho corresponding to monthly payment and its coefficient b1. Yes, and what is the standard error, the test statistic and the corresponding p value? We notice that the standard error is 0. 143 and if we divide the coefficient b 1 which is 1. 44 by the standard error of 0. 143, we get a value that is close to 10. So, in this case, this implies that the t statistic with

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497 degrees of freedom has a value of 10. 02. For this t statistic, we compute the p value. And it comes out to be fairly close to 0. And what does that imply? It implies that we can reject the null hypothesis confirming the linear relationship between monthly payment x 1 and the household debt y. And we know that this relationship is significant.

Let us look at the outcome of the hypothesis test with the null hypothesis of beta 2 equal to 0. For that we look at the next row corresponding to utilities and its coefficient B2. Yes, and we record the standard error, the test statistic and the p value.

We notice that the standard error in this case is 7. 87. Now, if we divide this coefficient B2, which is 168. 91 by the standard error of 7.87, we will get the t statistic with 497 degree of freedom to take a value of 21. 47. For this t statistic, the p value is very close to 0 and what does that imply again? It implies that we can reject the null hypothesis and it confirms that the linear relationship between utilities x 2 and that y is indeed significant.

Let us now look at the residuals. We have the residuals as well as the predicted debt given by our estimated regression equation. And what can we say about the standardized residuals in the output? There are a few values of the standardized residuals that are less than minus 2 and more than plus 2. By and large, they are within the range. Yes, and what does the plot of residuals against x1, which is the monthly payment, tell us? The plot of residuals against x1 is random, no pattern. And what about the plot of residuals against x2, utilities? That is also equally distributed above and below the x axis. Yes, and what about the plot of residuals against y hat, the predicted values of debt? Yes, that is also equally distributed above and below the x axis. Again, nothing to be concerned about.

What about the plot of the standardized residuals against y hat, the predicted values? It reveals the same random distribution above and below the x axis. There are a few values of standardized residuals that seem to lie above plus 2 and below minus 2. Yes, definitely. Yes, we expected that because we saw them in the table of the standardized residuals too. But overall, this seems to be a much better multiple linear regression model.

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MLR - Application with Examples - III

Was that helpful, Tejas and Ashwini? Clearly, that was a better model. I wonder what would happen if we added all three as independent variables. First income, monthly payment and utilities. Yes, all three variables seem to have the ability to explain debt in a household. How different is the analysis if we had three independent variables instead of only two?

Well, if you recall, our development of the theory was in terms of having up to p independent variables. So, we should be able to easily analyze the case with three independent variables. So, let us try that. We are now going to try a multiple linear regression model with three independent variables. We will use the household survey data from Manjula Naik.

We developed a multiple linear regression model with debt y as a function of two independent variables, namely monthly payment x1 and utilities as x2. The estimated regression equation was minus 140770 plus 1. 44 times the monthly payment plus 168. 9 times the utilities. The model had an adjusted r square of 0. 67. Both the F test for overall significance and the T test for individual significance showed that the assumptions behind multiple linear regression model hold. And that these linear relationships are significant. The outlier analysis also did not reveal any concerns. But we also saw that a multiple linear regression model with debt y as a function of the two independent variables, namely first income as x1 and monthly payments as x2 did well too.

The estimated regression equation in that case was 13081 plus. 0. 025 times the first income plus 2. 09 times the monthly payment. The model had an adjusted R square of 0. 45, not as good, but still okay. Both the F test for overall significance and the T test for individual significance in that case too showed us that the assumptions behind the multiple linear regression hold and that these linear relationships are significant. The outlier analysis also did not reveal any concerns. So, these observations therefore, suggest that perhaps a multiple linear regression model with three independent variables might perform well too. We will now develop such a multiple linear regression model with the dept y as the function of three independent variables. Namely, first income X1, monthly payment X2 and utilities as X3.

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Here I have the data in the spreadsheet with debt in column K, the first income in column H, monthly payment in column I and utilities in column J. Let us conduct the regression analysis using analysis tool pack and interpret the outputs. In analysis tool pack, we select the Y variable to be the debt data in column K.

And the X variables to be the first income in column H, monthly payment in column I and utilities in column J. And as we have done with our previous models, in addition to obtaining the output, let us also compute the residuals, the standardized residuals and the plots of these residuals. I will wait until you both get all the outputs.

Looks like you have them now. Let us look at the R squared. What did you get? We get a value of 0. 708 from ToolPak. Oh, good. That is even better than the 0. 67 we got with monthly payment and utilities as the independent variables. This is the best we have got so far. Yes, and we seem to have improved a lot from the simple linear regression models that had an R square of 0. 3 to 0. 35. This R square of 0. 70 implies that our multiple linear regression model with three independent variables is able to do much better. It is able to explain 70 percent of the variation in household debt as a function of the first income, the monthly payment and the utilities. What does the output tell us about the adjusted R square? The adjusted R square is pretty close to the R square value. It is also around 0. 70, just slightly lower than the R square that we computed. And we have been seeing this in all our models for this data set so far.

Next, let us look at the standard error. We see that the standard error is 18699, which is lower than the standard error of 19839 that we got with utilities and monthly payment as independent variables. And how does that compare with what we had for the two simple linear regressions? For the earlier simple linear regressions, we had standard errors that were more. Yes, we had standard errors of around 28, 000 in both cases. And having first income, monthly payment and utilities as the three independent variables has reduced the standard error and increased our R squared.

Let us now examine the output of the ANOVA analysis. The table shows that SSR is 4. 21 times 10 raised to the power 11 and SSE is equal to 1. 73 times 10 to the power 11. The SSR is even better than the value of 3. 99 times 10 to the power 11 that we got for our previous model. The SSE is lower than the 1. 96 times 10 to the power 11 that we got with the previous

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model. And what happened to the SST? The SST remains at 5. 94 into 10 raised 11. It should, right? Because the total variation in debt is independent of the number of independent variables we choose to explain it. Yes, and the higher SSR in this case is saying that the multiple linear regression with three independent variables is able to explain more of the variation in debt than the other two choices.

And of course, They do far better than the simple linear regression models. Now let us look at MSR and MSE. MSR is 1. 4 times 10 to the power 11 and MSE is 349653451. When we had only two independent variables, the MSR was 1. 99 times 10 to the power 11 and the MSE was 393601978. Both are better when we have three independent variables.

And since we are looking at a multiple linear regression with three independent variables now, the degree of freedom for the mean square regression is equal to 3. And the degree of freedom for the mean square error is now equal to 496, okay. What does the F statistic come out to be in this case? The F statistic is the ratio of MSR over MSE and that comes out to be 401. 6. Yes, and what does that mean in terms of our p value? It is close to 0. Yes, and therefore, We can conclude that this allows us to reject the null hypothesis that beta 1, beta 2 and beta 3 are all equal to 0. And that implies that there is enough evidence to claim that all of these 3 cannot be equal to 0.

Therefore, confirming a significant overall linear relationship between debt, first income, monthly payment and utilities. Next, let us look at the table with the coefficients. What are the values of B0, B1, B2 and B3? B0 is minus 140378. B1 is 0. 017 for first income. B2 is 0. 96 for monthly payment. And B3 is 157. 95 for utilities. Yes. And how would we interpret the coefficient B1? For every rupee increase in first income, both monthly payment and utility is being held constant. The debt would increase by 0. 017. And how do we interpret the coefficient B2? For every rupee increase in monthly payment, for both first income and utilities being held constant, the debt increases by 0. 96. And finally, how do you interpret the coefficient B3? For every rupee increase in utilities, both first income and monthly payment being held constant, the debt increases by 157. 95. Then the next question, are these coefficients significant? Could any of these betas, beta 1, beta 2 or beta 3 be equal to 0?

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What will be the outcome of the hypothesis test with beta 1 equal to 0? We need to look at the row corresponding to first income and its coefficient b1. Yes, and we look at the standard error, the test statistic and the p value like we have done before. We notice that the standard error is 0. 002 and if we divide the coefficient b1 which is 0. 017 in this case by the standard error of 0. 002 we get a value close to 7. 96. This is the value of the test statistic which is a t random variable with 496 degrees of freedom. For this test statistic what is the p value? It is close to zero. Yes, and what does that imply? It implies that we can reject the null hypothesis confirming that the linear relationship between first income x1 and debt y is indeed significant.

Now we do the same with the other coefficients. What is the outcome of the hypothesis test with a null hypothesis of beta 2 equal to zero? For that, we look at the next row corresponding to monthly payment and its coefficient B2. Yes, and we look at the standard error, the test statistic, the p value as we have always done.

We notice that the standard error is 0. 148 and if we divide the coefficient B2, which is 0. 96 in this case, by the standard error of 0. 148, we get a t statistic with 496 degrees of freedom to take a value of 6. 5. And for this t statistic, the p value is close to 0. That implies that we can reject the null hypothesis and confirm that the linear relationship between monthly payment x2 and debt y is indeed significant.

Finally, let's look at the outcome of the hypothesis test with the null hypothesis of beta 3 being equal to 0. For that, we look at the next row corresponding to monthly payment and its coefficient b3. Now, we look at the standard error, test statistic and p value. That's right. And we notice that the standard error in this case is 7. 54. And if we divide the coefficient b3, which is 157. 95 by the standard error of 7. 54, So, we get a t statistic with 496 degrees of freedom to take a value of 20. 95. For this value of the t statistic, the p value is close to 0. So again, we can reject the null hypothesis and it confirms that the linear relationship between utility is X3 and debt Y is indeed significant.

Good. Now, let us look at the residuals. We have the residuals as well as the predicted debt given by our estimated regression equation. What can we say about the standardized residuals? By and large, they are within the range. Yes. And what does the plot of residuals against x1, which is our first income, tell us?

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The plot of residuals against x1 is random, both positive and negative values. And what does the plot of residuals against x2, monthly payment, tell us? That is also equally distributed above and below the x axis. Yes, That is true. And what about the plot of residuals against x3 utilities. These residuals have a random distribution, both above and below the x axis. Yes, and what about the plot of residuals against y hat, the predicted value of depth?

That is also equally distributed above and below the x axis again nothing to be concerned about. And what about the plot of the standardized residuals against y hat, the predicted values? It reveals the same random pattern with points above and below the x axis. There are a few values of standardized residuals that seem to lie above plus 3 and below minus 3.

Why are we seeing these now? Yes, since our estimated regression equation is giving a better fit. And the standard error is lower. We are seeing more of such points. These could be outliers. But overall, this seems to be a much better multiple linear regression model. See, that was not much different at all, is it not, Tejas and Ashwini?

Yes, definitely. When Excel does all the hard work of computations. But that's okay. It helps us spend more time on interpretation. Shall we do a similar analysis, but on a different data set? It might teach us something. Sure, let's do that.

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MLR - Application with Examples – IV

Let us look at a data set that we had seen in module 1. It was prepared by Vijay Gowda, who is a real estate analysis working for Magic Bricks. He is analyzing the property prices for flats in Jayalakshmi Puram in Mysuru to determine the market rates for comparable properties. His data set had 30 observations with variables such as selling price in lakhs, the square feet area, the number of bedrooms, the number of bathrooms in the apartment and information about whether the property was located in a premium locality, a gated complex and if it had good amenities.

Let us see if we can develop a linear regression model that is able to explain the selling price of these properties in terms of these variables. We will first develop a simple linear regression model with selling price y as a function of one independent variable, namely the area in square feet, x. Shall we draw a scatter plot first? Sure, why not? Let us do that. I get a plot that shows a rough linear trend, but several outlier possibilities. Yes, and what is the trend line equation and the associated r square.

I get the equation to be 28. 37 plus 0. 0968 times x, the area in square feet. Yes, and the r square is only 0. 44. Okay, this should be interesting. I am sure we will learn something about selling price of flats. Let us do the full regression analysis using analysis tool pack. Here, I have the data in the spreadsheet with the selling price in column H and the area square foot in column G.

In analysis tool pack, we select the y variable to be the selling price data in column H and the x variable to be the area in square foot in column G. As we have done with our previous models, in addition to obtaining the outputs, let us also compute the residuals the standardized residuals and their plots.

Do you have all these outputs? Looks like you have them now. Okay, let us look at r square. What did you get? We get a value of 0. 44 from tool pack also. Yes, and that is consistent with our trend line. This r square of 0. 44 implies that our simple linear regression model with area square foot as the only independent variable explains only 44 percent of the variation in selling prices observed in the 30 data points.

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Now what does the output tell us about adjusted R square? The adjusted R square is pretty close to the R square about 0. 43 and we have been seeing this in all our models so far. Let us now look at the standard error. We see that the standard error is 70. 57. This is an estimate of the sigma, the standard deviation of the error epsilon in our model.

Next, let us examine the ANOVA output. The table shows us an SSR of 111539 and an SSE to be equal to 139462. And what is the value of SST in this case? The SST is 251001. As expected, SSR is about 44 percent of the SST. Yes, and the regression is explaining about 44 percent of the variation.

Now, let us look at MSR and MSE. In this case, MSR is 111539 and MSE is 4981. Since we have only one independent variable, the MSR is equal to the SSR and is equal to 111539. For the MSE, the degree of freedom is 28, and so we have SSE divided by 28 to be equal to 4981. Okay. What does the F statistic come out to be? The F statistic is the ratio of MSR over MC and that comes out to be 22. 39. Yes. And what does that mean in terms of the p value? It is close to zero. Yes. And therefore, we conclude that this allows us to reject the null hypothesis that beta one is equal to zero. And that implies that there is enough evidence to claim that there is a significant linear relationship between the selling price and the area in square feet.

Hardly surprising, isn't it? But definitely reassuring. Next, let us look at the table with the coefficients. What are b 0 and b 1? b 0 is 28.37, b 1 is 0.09 for area. Yes, and how do we interpret this coefficient b 1? For every square foot increase, the selling price increases by 0. 09 lakhs or 90, 000. That seems high. Yes, it is. And that is because we have only one independent variable in this model. We will address that later.

What is the outcome of the hypothesis test with the null hypothesis of beta 1 equal to 0? We need to look at the row corresponding to area square feet and its coefficient b1. Yes, and we look at the standard error, the test statistic, and the p value as we have always done.

We notice that in this case, the standard error is 0. 02. And if we divide the coefficient b1, which is 0. 096 by the standard error of 0. 02, we get a value that is close to 4. 73. So, in this case, it implies that a t statistic with 28 degrees of freedom has taken a value of 4. 73. And from this t statistic, what is the p value? It is close to 0. Yes. And what does that tell us? It tells us

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that we can reject the null hypothesis confirming that a linear relationship between selling price y and the square foot x is indeed significant.

Let us now look at the residuals. We have the residuals as well as the predicted selling price given by our estimated regression equation. What can we say about these standardized residuals? Some of the residuals are high. Yes, and what does the plot of the residuals against x, which is the selling price, tell us? The plot of residuals against x is random, both positive and negative values. Yes, and what about the plot of residuals against y hat, the predicted value of the selling price?

That is also equally distributed above and below the x axis. Yes, again, nothing to be really concerned about. And what about the plot of the standardized residuals against y hat, the predicted selling price? There are a few values of standardized residuals that seem to lie above plus 3 and below minus 3.

Yes, these could be outliers, but this is a good start. Now we can see if we could improve this model by building a multiple linear regression model. Yes, let's do that.

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MLR - Application with Examples – V

Let us continue to work with the data set prepared by Vijay Gowda, who is a real estate analysis working for Magic Bricks. He wants to analyze the property prices for flats in Jayalakshmi Puram in Mysore to determine the market rates for comparable properties. As we just saw, his data had 30 observations with variables such as selling price in lakhs, square feet area, number of bedrooms, number of bathrooms in the apartment and so on.

It also had information about the property. It also had information about whether the property was located in a premium locality, a gated complex and if it had good amenities. We developed a simple linear regression model. with the selling price y as a function of one independent variable, namely the area in square feet x.

We obtained the estimated regression equation to be 28. 37 plus 0. 0968 times x, the area in square feet. However, this model gave us an r square of 0. 44. This r square of 0. 44 implies that the simple linear regression model with area square foot as the only independent variable explains only 44 percent of the variation in selling prices observed in the 30 data points.

Both the F test for significance and the T test confirmed the presence of a significant linear relationship between area and the selling price. Now, we want to see If we can use a multiple linear regression model with other quantitative variables to try and improve this model. We want to see if including additional quantitative variables will allow us to build a model that explains a higher percentage of the variation in selling price of flats in this data set.

What quantitative variables do we have in this data set? The number of bedrooms, the number of bathrooms in the flat. Yes, and any others? Premium locality, community complex and amenities. And why do you say that they are quantitative variables? They take numeric values of 0 and 1. Why is that not true? No, Tejas, they are in fact qualitative variables.

Do you remember what kind of qualitative variables they are? Are they qualitative variables but of numeric type measured on a scale? Yes Ashwini, that is right. They are qualitative variables and on a nominal scale, just indicating how the data falls into certain categories. There

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is no ordering associated with it. We are not saying that the data labeled with 0 is better than a data labeled with 1, okay.

So, we have choices of additional variables. And the choices we have are the number of bedrooms and the number of bathrooms. Let us build a multiple linear regression model with the selling price y as a function of two independent variables, namely the area square foot as x1 and the number of bedrooms as x2.

Let us do the full regression analysis using analysis tool pack. Here, I have the data in a spreadsheet with the selling price in column H and the area square foot in column E and the number of bedrooms in column F. In the analysis tool pack, we select the Y variable to be the selling price data in column H and the X variables to be the area square foot and the number of bedrooms in columns E and F.

And as we have done with our previous models, let us obtain the output, compute the residuals, the standardized residuals and the plots of the residuals. I will give you both a minute to do that and let me know when you are done. Are you done? Okay. Let us now look at the R square. What did you get? We get a value of 5464 from toolpath. That is not impressive. It was 0. 44 for a simple linear regression. Yes, that is disappointing. But the outputs will tell us why it did not improve. The good part of these outputs is that when the R square is not satisfactory, the diagnostics can also be used to find what went wrong and what could improve the model.

So, in that sense, this might be a good learning opportunity. Let us continue with our analysis. Now, what does the output tell us about the adjusted R square? The adjusted R square is pretty close to the R square about 0. 42, nothing surprising there. Next, let us look at the standard error. We see that the standard error is 70. 74, fairly close to the 70. 57 that we got with simple linear regression. And as we know, this is an estimate of the sigma, the standard deviation of the error epsilon in our model. Next, let us move on to the ANOVA output. Of course, The sum of squares total SST is 251001. That should not change from our simple linear regression model.

The table shows that SSR is 115892. For the simple linear regression with only area square foot as the independent variable, it was 111539. So adding the second variable did help to some

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extent. Of course, SSR divided by SST is the R square that we got, the 0. 46 or the 46 percent. We also see that the sum of squares error SSE is equal to 135109, which is marginally lower than the 139461 that we got with a simple linear regression.

Now, Let us look at MSR and MSE. What do we have? We see that the MSR is 5 7 9 4 6, and MSE is 5 0 0 4. Since we have two independent variables, the MSR is equal to SSR divided by two and is equal to 5 7 9 4 6. For the MSE, the degree of freedom is 27, and so we have the sum of squares error, SSE of 1 3 5, 1 0 9 divided by 27 to be equal to 5004. Okay, what does the F statistic come out to be? We know that the F statistic is the ratio of MSR over MSE and in this case that comes out to be 11. 58. And what does that mean in terms of our p value? It is 0. 0002 Yes, and so therefore, we can conclude that this allows us to reject the null hypothesis that both beta 1 and beta 2 are equal to 0.

Therefore confirming a significant overall linear relationship between selling price and area square foot and the number of bedrooms. Again, not surprising, but definitely reassuring. Next, let us look at the table with the coefficients. So, what are b 0, b 1 and b 2? b 0 is minus 29.19, b 1 is 0.085 for area and b 2 is 23. 46 for number of bedrooms. Yes. So, the coefficient b 1 did not change much from the 0. 09. It was for the simple linear regression that we did. What is the outcome of the hypothesis test with the null hypothesis of beta 1 equal to 0? We need to look at the row corresponding to area square fit and its coefficient b 1.

Yes, and we look at the standard error, the test statistic and the p value. We notice that the standard error is 0. 024 and if we divide the coefficient b 1 which is 0. 085 in this case. by the standard error of 0. 024, we get a value of 3. 46. In this case, this implies that it is a t statistic with 27 degrees of freedom and it takes a value of 3. 46. For this t statistic, what do we get as the p value? It is 0. 002. Yes, and what does that imply? It implies that we can reject the null hypothesis confirming that a linear relationship between selling price y and area square foot x1 is indeed significant. Okay, now what is the outcome of the hypothesis test with the null hypothesis of beta 2 equal to 0?

We need to look at the row corresponding to number of bedrooms and its coefficient b1. Yes, and we look at the standard error, the test statistic and the p value. We notice that the standard error is 25. 15 and if we divide the coefficient b1 which is 23. 46 in this case by the standard

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error of 25. 15 we get a value of 0. 932. So in this case this implies that the t statistic with 27 degrees of freedom takes a value of 0. 932. So for this t statistic what is the p value? Oh, the p value is 0. 36, so we cannot reject the analysis. Yes, and what does it imply? It means that the coefficient beta 2 could be equal to zero. But that does not make sense. Selling prices should be associated with the number of bedrooms in the flat. True. So, why is this happening? Is this because of the multicollinearity, the correlation between area square feet and the number bedrooms might be significant. Quiet possible. Yes, that is what must be happening. It would make sense then. Yes. So we will investigate this a little bit further.

But let us quickly look at the residuals and the residual plots before we do that. What can we say about the standardized residuals? There are two values above 2, but otherwise they are all looking ok. And what does the plot of residuals against x1 and x2, the area square foot and the number of bedrooms, tell us?

Both plots have a nice distribution about the x axis. No particular patterns. And what about the plot of residuals against y hat, the predicted values of the selling price? That is also equally distributed above and below the x axis. Again nothing to be concerned about. And what about plot of standardized residual against y hat, the predicted selling price. As expected, two values above two and the others are okay. Yes, so now let us investigate the result that we got for the individual significance a liite bit more. What should we do? We should compute the correlation coefficient between area square feet and number of bedrooms. Maybe it is more than 0. 7. Let us do that. Actually, let us compute the correlation coefficient between selling price, area square feet, number of bedrooms, and number of bathrooms.

Let's go back to the analysis tool pack, select all those columns as inputs, and get the correlation table. What do you get? We are looking at the correlation between area square feet and number of bedrooms, right? It is 0. 542. That is not too high. You say it only if it is above 0. 7. Yes, just as a thumb rule, just as a rule of thumb. That is why we need to do the diagnostic and not just conclude. Okay so is it multicollinearity that is causing high P value.

Yes, it is and as Tejas said, it is to be expected that the area square foot and the number of bedrooms are highly correlated. So what the regression diagnostics is telling us is that given

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that we already have area square feet as an independent variable in the model, adding the number of bedrooms as a second independent variable is not adding much value.

So, we might as well remove it from the model. Okay, if we remove that, the only other quantitative variable we can add as x2 to try and improve the R square of the model is number of bathrooms. Yes, that is correct. So should we add that instead? But number of bathrooms should also be correlated with number of bedrooms an area square feet. In fact, the co-relation between area square feet and number of bathrooms is 0.659. Yes. In fact, it is higher than the correlation we had between area square foot and the number of bedrooms. So if we add the number of bathrooms instead of the number of bedrooms as the second independent variable, are we likely to see multicollinearity there too? Very likely. Should we investigate? Yes.

Let us double check if we do see the effect of multicollinearity. Okay. So let us build a multiple linear regression model with the selling price y as a function of the two independent variables, namely the area square feet as x1 and the number of bathrooms now as x2. Let us do the full regression analysis using analysis tool pack.

Here I have the data in a spreadsheet. In the analysis tool pack, we select y to be the selling price data and the x variables to be the area square foot and the number of bathrooms. I will give you both some time to do that. Are you done? Okay. Let us look at the R square value. What did you get? We get a value of 0. 45 from toolpath, not very different. Yes, now we are not very surprised with this value. Let us continue with our analysis. What does the output tell us about the adjusted R square? The adjusted R square is pretty close to the R square, about 0. 41. Next, let us look at the standard error. We see that the standard error is 71. 18 slightly more than the 70. 74 that we got with number of bedrooms as x2. Let us now move on to the ANOVA output. Of course, the sum of squares total SST is still 251001 that should not change from the simple linear regression. The table shows that SSR is 114185 which is similar to the 115892 that we got when the number of bedrooms was x2.

We also see that SSE is equal to 136816, similar to the 135109 that we got before. Now let us look at MSR and MSE. We see that the MSR is 57092 and MSE is 5067, also similar. Yes, and so what does the F statistic come out to be? Well, we know that the F statistic is the ratio of MSR over MSE. And in this case, that comes out to be 11. 26. And what does that mean in

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terms of our p value? It is 0. 0003 Yes. And so therefore, we conclude that this allows us to reject the null hypothesis that both beta 1 and beta 2 are equal to 0. Therefore confirming a significant overall linear relationship between the selling price and area square feet and the number of bathrooms.

Next, let us look at the table with the coefficients. What are the values of B0, B1 and B2? B0 is 10. 61, B1 is 0. 084 for area and B2 is 17. 80 for number of bathrooms. Yes, and what is the outcome of our hypothesis test with a null hypothesis of beta 1 equal to 0? We need to look at rocus finding the square feet and its coefficient b1.

Yes, and we look at the standard error, the test statistic and the p value. We notice that the p value is 0. 005. So that implies that we can reject the null hypothesis, confirming that the linear relationship between selling price y and the area square fit x1 is indeed significant.

Okay. Now, what is the outcome of the hypothesis test, where the null hypothesis is beta 2 is equal to 0? We need to look at the row corresponding to number of bathrooms and its coefficient b2. Yes, and we look at the standard error, the test statistic and the p value. The test statistic t is 0. 723. And for this t statistic, what do we get to be the p value? Yes, the p value is 0. 48. So we cannot reject the analysis.

Yes, and what does that imply? It implies that beta 2 could be equal to zero. Okay, so it is multicolinearity again, showing up in the diagnostics. Yes, it is indeed. But why does the F test cause any problems? Yes that is a good point. The F test is a test for overall significance. And what is the null hypothesis there in that F test?

The null hypothesis is that both beta 1 and beta 2 are equal to 0. Yes, and our individual t tests are saying that beta 1 is not equal to 0. That is the area square foot matters. So the F test for overall significance says that there is a linear relationship. So that means we are okay trying to fit a multiple linear regression.

But the individual t test of significance is telling us that we have to work on finding the correct independent variables to include in the model. Adding the number of bedrooms or the number

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of bathrooms is not going to yield significant improvement. I see. How the combination of these tests is telling us some useful information.

Yes, the diagnostics are very helpful both when the model is acceptable and when the model needs improvement. Okay, I agree with all this, but we have ran out of variables to add into the regression. Now we have only qualitative. Yes, that is true. Is it possible to add qualitative variables into multiple linear regression?

Yes, it is possible. And that is going to be the focus of our next module. So you will need to wait for the next module to see how we do it. And at that time, we will revisit this example.

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Intercept and OB

So, let me summarize what we have learnt in this module before we conclude here. I have a question that has been bothering me for a while. Yes, Ashwini, what is it? In all our regression analysis, we have been analyzing the outputs for the independent variables b1, b2, and so on. We never analyze the output corresponding to the intercept b0.

Why is that? That's a very good question, Ashwini. As you know, when we get outputs through analysis tool pack, the outputs for both simple linear regression and multiple linear regression, it usually contains results for a hypothesis done on beta 0 as well. It corresponds to a null hypothesis of H naught given by beta 0 is equal to 0.

This implies that we are testing whether the intercept parameter is equal to 0 or not. Inferences on the intercept parameter beta 0 are special cases of inferences on the fitted value of the regression line as a whole. That is, establishing confidence intervals for both the predicted and the estimated values of the dependent variables.

We have not gone into details of those in this module. And also, in most cases, the null hypothesis of H naught given by beta 0 is equal to 0 is not of particular interest. And its p value given in the computer output is usually not that important. That is the reason we have not discussed the beta 0 or the b 0 coefficient in detail.

If the training data set that we are using does not include the points where the independent variables take a value of zero, in that case, if we use the estimated regression equation to make predictions on the dependent variable, when the independent variable takes a value of zero, we would be doing an extrapolation of what the training set implies.

And as we have said earlier, it is really important to use the regression equation for prediction only within the ranges of the independent variables that are in the training dataset. It is risky to use the estimated regression equation to make prediction for values of independent variables that lie outside the range of variables included in the training dataset.

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Now, does that help, Ashwini? Yes, it does. Thanks. Okay, let us talk a little bit about a concept known as omitted variable bias that we see in multiple linear regression and in simple linear regression. Let us consider what happened in our example of the household data, where we were trying to predict the debt y as a function of first income x1, monthly payments x2 and utilities x3.

If you recall, we had done a simple linear regression with debt Y as a function of just the first income. We saw that this model gave us an R square of only 0. 31 or 31 percent.

Subsequently, we improved the model in steps to finally lead to a multiple linear regression model that expressed debt Y as a function of first income X1, monthly payment X2, and utilities X3. We saw that this multiple linear regression model gave us an R square of 0. 71. When we did the simple linear regression of debt Y as a function of only first income, we were ignoring the other variables that we found to be relevant subsequently.

And this is what led to a low R square in the simple linear regression model. In general, such omissions can result in the coefficient for the independent variable, first income in this case, to be very different from their true contributions. This is because the included independent variable, first income in this case, tries to also model the effect of the ignored independent variables onto itself.

This phenomenon leads to a bias in the model known as the omitted variable bias. As part of the regression analysis, we can actually quantify how much an independent variable overestimates or underestimates its true contribution in the presence of an omitted variable bias. However, we will not get into the details of this process in this module, but we will talk a little bit about how to identify whether this phenomenon is happening or not in the model.

For omitted variable bias to exist, at least one of the following must be true in the data. The omitted variable must be correlated to the dependent variable. In the context of our housing data, if we are omitting monthly payment and utility, and only including first income in our simple linear regression model.

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That is, we are trying to predict the debt Y as a function of only the first income X, then the omitted variable bias could occur if debt Y is correlated with either monthly payment or utilities. Can you compute these sample correlations using analysis tool pack? What do you get? We see that debt has a sample correlation of 0. 56, 0. 61, and 0. 78, with first income, monthly payment, and utilities, respectively.

Yes, so omitted variable bias can occur in this case. Additionally, omitted variable bias might also occur if the omitted variable is correlated to some of the independent variables. That means, in the context of our example, omitted variable bias might occur if the first income is correlated with either monthly payment or utilities.

And we know looking at the correlation table that is indeed the case. We can see that monthly payment and first income had a correlation of .55.. Yes, and we

also see that utilities and first income have a correlation of 0. 39. Yes, and utilities and monthly payment have a sample correlation of 0. 49. Yes, so additional reasons why omitted variable bias might occur in our case. If we consider a simple linear regression where debt is explained only as a function of the independent variable first income, then we could argue that we are ignoring the other independent variables such as monthly payment and utility.

And depending on the effect of the omitted variables x2 and x3, which are monthly payment and utilities in this case, on the dependent variable y, which is debt in this case, the included variable x1, which is first income in this case, the effect of the omitted variable bias can be explained. Let us talk a little bit about how do you detect if omitted variable bias is occurring in your model.

If omitted variable bias is occurring, you would see a significant increase in the R square when you include the additional variables that were omitted in the first place. If you recall, our R square increased from 0. 31 to 0. 71. We should also see a significant drop in the standard error of this model when the additional in the first place are subsequently included into the model.

You can see the signs of omitted variable bias by looking at the standard errors and the p values of the multiple linear regression model that has the additional variables included compared to

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the standard errors and the p values of a regression model that omits certain variables. You will see a significant drop in the p value when the necessary variables are included into the model, along with a drop in the standard error of the independent variables.

We will limit our discussion of omitted variable bias to this level of detail for now.

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MLR – Recap

Let us now summarize what we have done before we close this module. We began with a basic idea behind multiple linear regression. Essentially, it is an extension of simple linear regression in the sense that we are trying to establish a linear relationship between a dependent variable y and several independent variables x.

We saw the basic multiple regression model to be equal to y is equal to beta 0 plus beta 1 times x 1 plus beta 2 times x 2 and so on up to beta p times x p plus epsilon. The expected value of y, E of y was then equal to beta 0 plus beta 1 times x 1 plus beta 2 times x 2 and so on up to beta p times x p.

Then, w e discussed the steps that are typically used to build and validate and implement a multiple linear regression model. We then used the sample data and the least square method to develop the estimated regression equation y hat is equal to b 0 plus b 1 times x 1 plus b 2 times x 2 and so on up to b p times x p.

These coefficients b 0, b 1, b 2 and so on up to b p are sample statistics that are used to estimate the unknown population parameters, beta 0, beta 1 and so on up to beta p in the population. To understand the performance of the model and validate it, we undertook several steps. We saw the generalized version of the coefficient of determination known as the multiple coefficient of determination.

We also discussed the assumptions behind the error term epsilon in the model. We then did the F test for overall significance of the linear relationships and then the T test for the individual parameters beta 1, beta 2 and so on up to beta p. To test for the significance of the individual parameters in the model, we also saw concepts of multicollinearity and omitted variable bias.

What else did we cover? We were able to analyze several examples using analysis tool pack and in this process of comparing and improving the models, we learned about model building. Yes, and we learned that it is indeed an iterative process. We also saw how we can analyze residuals to detect outliers and influential observations.

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Yes, that is true. So, we have progressed even more in our journey on inference where we are using data from a single sample to make strong inferences about the population and support our decision making. In the next module, we will see how to extend multiple linear regression to include categorical variables.

And then we will wrap up our course with a summary. Thank you Ashwini and Tejas for all your help and support. And thank you all for your attention. I will see you next time.

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